Experiment 3

WAVELENGTH DIVISION MULTIPLEXING (COARSE WDM and DENSE WDM)

Design & simulate 8 × 2 . 5 Gbps WDM system and analyse the BER Performance with the following parameters

Design Specification:

Bit Rate- 2.5 Gbps

Line Encoding: NRZ

Modulation: External Modulation

Experimentation

1. Plot the spectrum of the transmitter(Multiplexed Spectrum) and receiver comment on the result.
2. CWDM
Channel Spacing= 20 nm
3.DWDM
Channel Spacing : 0.8nm
4. Write your inference on the result obtained.

Aim:

To design Coarse and Dense WDM Systems and to analyze the BER performance.

Software Required:

Optisystem 22.0 VERSION

Theory:

WDM: Wavelength-Division Multiplexing is now one of the most widely used technology for high-capacity optical communication

systems. Figure:1 schematically shows a typical WDM transmission system. At the transmitter side, multiple optical transmitters – each emitting at a different wavelength – individually send signals and these signals are multiplexed by a wavelength multiplexer (MUX). The multiplexed signals are then transmitted over one main transmission line (optical fiber cable). At the receiver side, the signals are de-multiplexed by a wavelength de-multiplexer (DEMUX) and sent to multiple receivers.

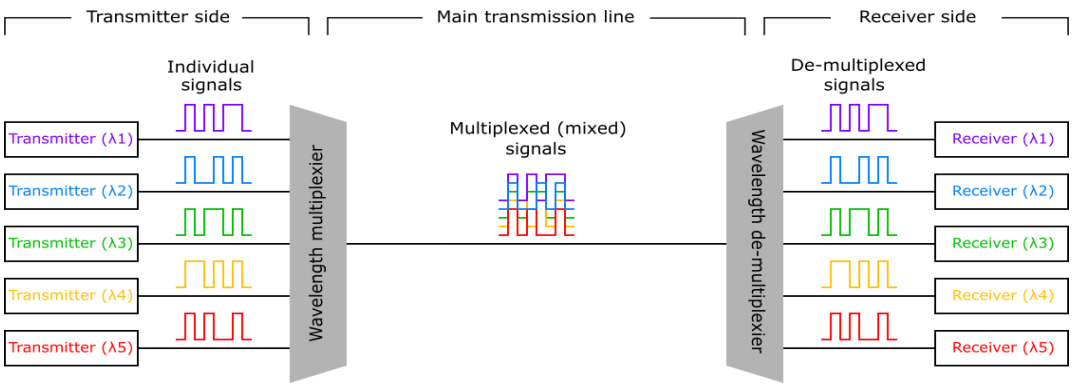


Figure:1

CWDM (Coarse WDM)

The need for a tight channel spacing in DWDM technology mainly arises from the relatively narrow gain bandwidth of EDFA (compared to the entire optical telecommunication bands). On the other hand, if the transmission distance is less than 100 km and no amplifiers are needed, a wider channel spacing can be an option.

A wider channel spacing allows the use of inexpensive components such as:

- Uncooled transmitter laser diode (LD) with a large wavelength variation
- MUX and DEMUX with a relaxed channel spacing

and as a result, the total cost for installation and operation becomes less expensive. Such WDM systems are called coarse WDM (CWDM), and ITU-T G.694.2 defines one wavelengths allocation for CWDM systems. There are 18 center wavelengths with 20 nm spacing from 1271 nm to 1611 nm, covering the O-, E-, S-, C- and L-bands. All the 18 wavelengths are not necessarily be used.

Dense WDM (DWDM)

In order to transmit optical signals over a long distance (> 100 km), optical fiber amplifiers are needed to compensate the loss of an optical fiber. As the gain bandwidth of an optical fiber amplifier is rather limited, tight wavelength spacing is needed to put a large number of channels into the gain bandwidth. The dense WDM (DWDM) technology has been developed for long distance transmission systems, fully utilizing the gain bandwidth of erbium-doped fiber amplifier (EDFA). EDFA has optical gain in the C-band and L-band and for example, a total of 115 wavelength channels are transmitted in one fiber with 100-GHz (~0.8 nm) frequency spacing. Several different frequency spacings for DWDM applications are defined in ITU-T G.694.1, and an appropriate spacing is chosen depending on system requirements (total capacity, bit rate per channel, distance, etc.).

Use of WDM technology in telecom network

Telecom networks are roughly classified into three categories, the core, metro, and access networks . The core network connects major cities (>100 km) and DWDM technology is often used. The metro network is used inside a metropolitan area, typically 50~80 km, and CWDM may be used. The access network rarely uses WDM technology at present, as the requirement for transmission capacity is much less than the core and metro networks.

CALCULATION:

DWDM:

Channel Spacing : 0.8nm

User Bandwidth: 10 GHz

Bit Rate- 2.5 Gbps

Line Encoding: NRZ

Modulation: External

Channel Spacing= 0.8 nm

CALCULATION:

λ_1= 1550 nm

λ_2= λ_1 nm + 0.8 nm=

λ_3= λ_2 nm+0.8 nm=

λ_4= λ_3 nm+ 0.8 nm=

λ_5= λ_4 nm+ 0.8 nm =

λ_6= λ_5 nm+ 0.8 nm=

λ_7= λ_6 nm+ 0.8 nm=

λ_8= λ_7 nm+ 0.8 nm=

DWDM:

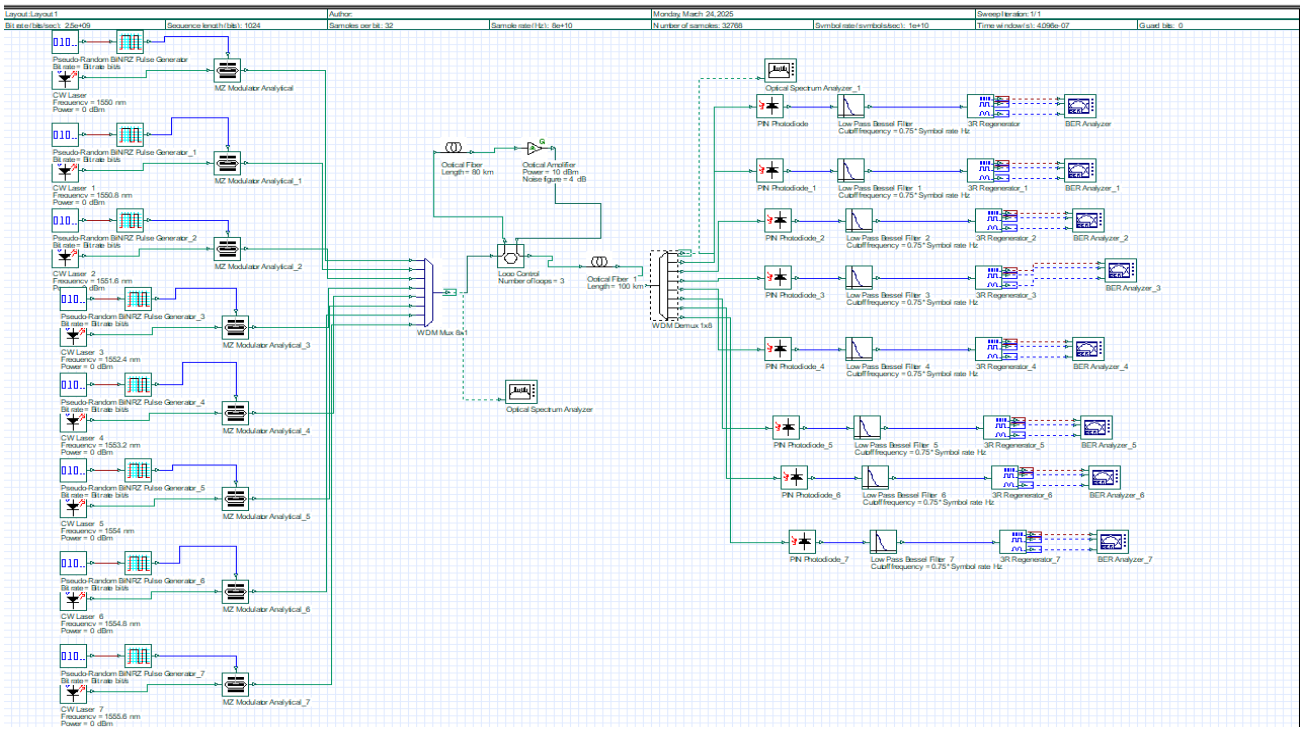


Table:DWDM

	Min. BER	Max. Q
1	6.33096e-14	7.404

	Min. BER	Max. Q
1	5.24543e-15	7.7291

	Min. BER	Max. Q
1	3.08506e-13	7.1915

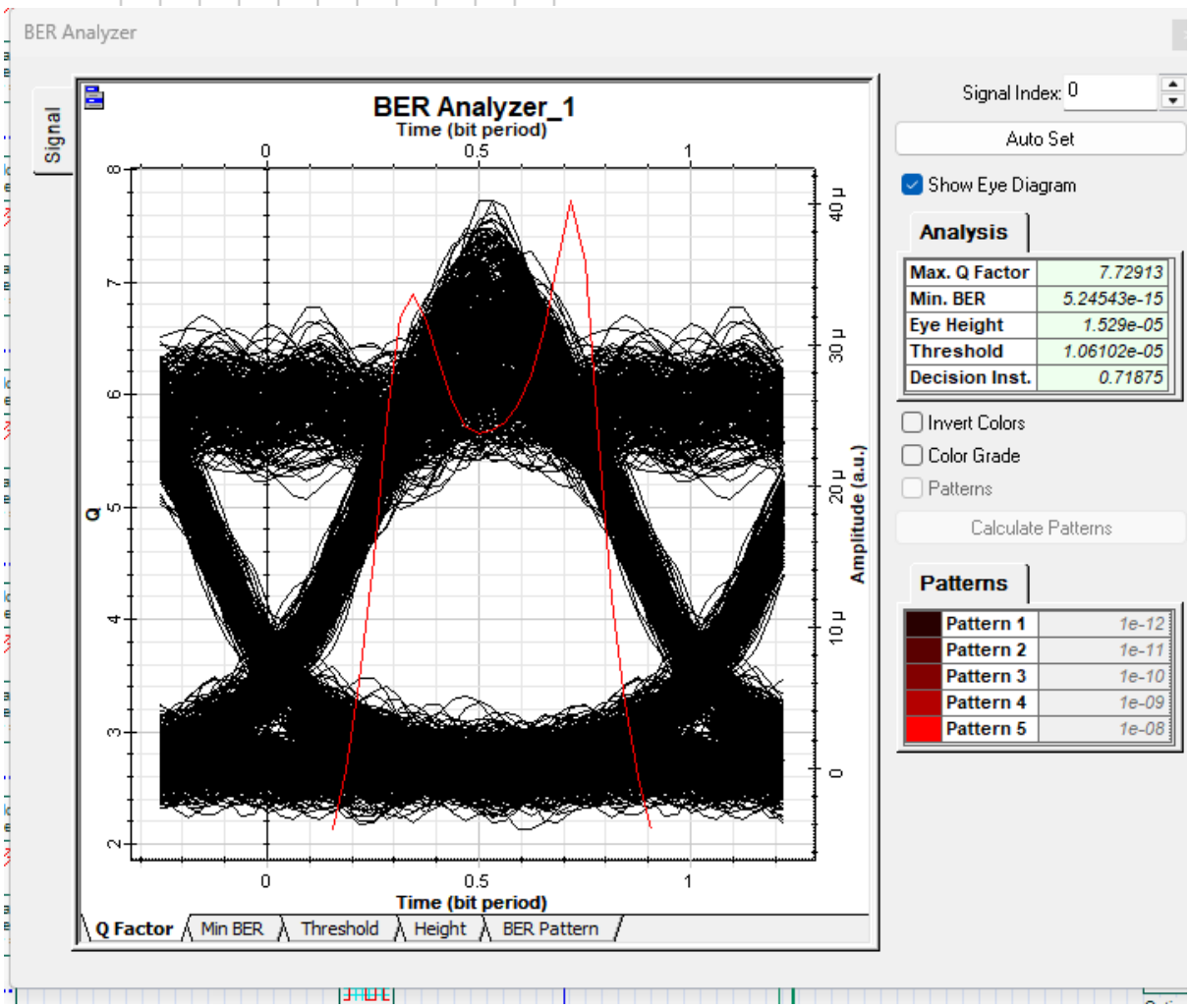
	Min. BER	Max. Q
1	7.45744e-14	7.3834

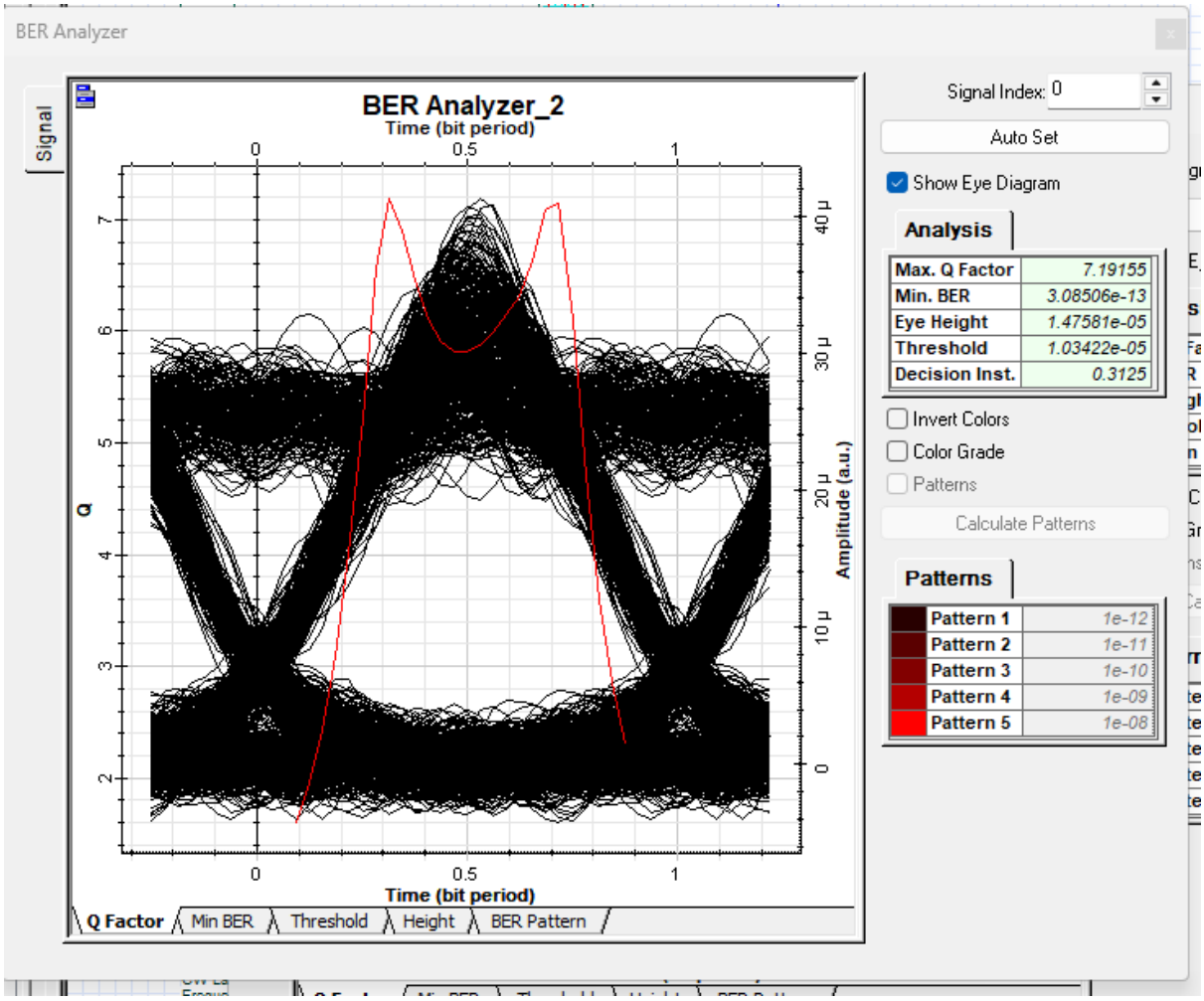
	Min. BER	Max. Q
1	1.38133e-13	7.2997

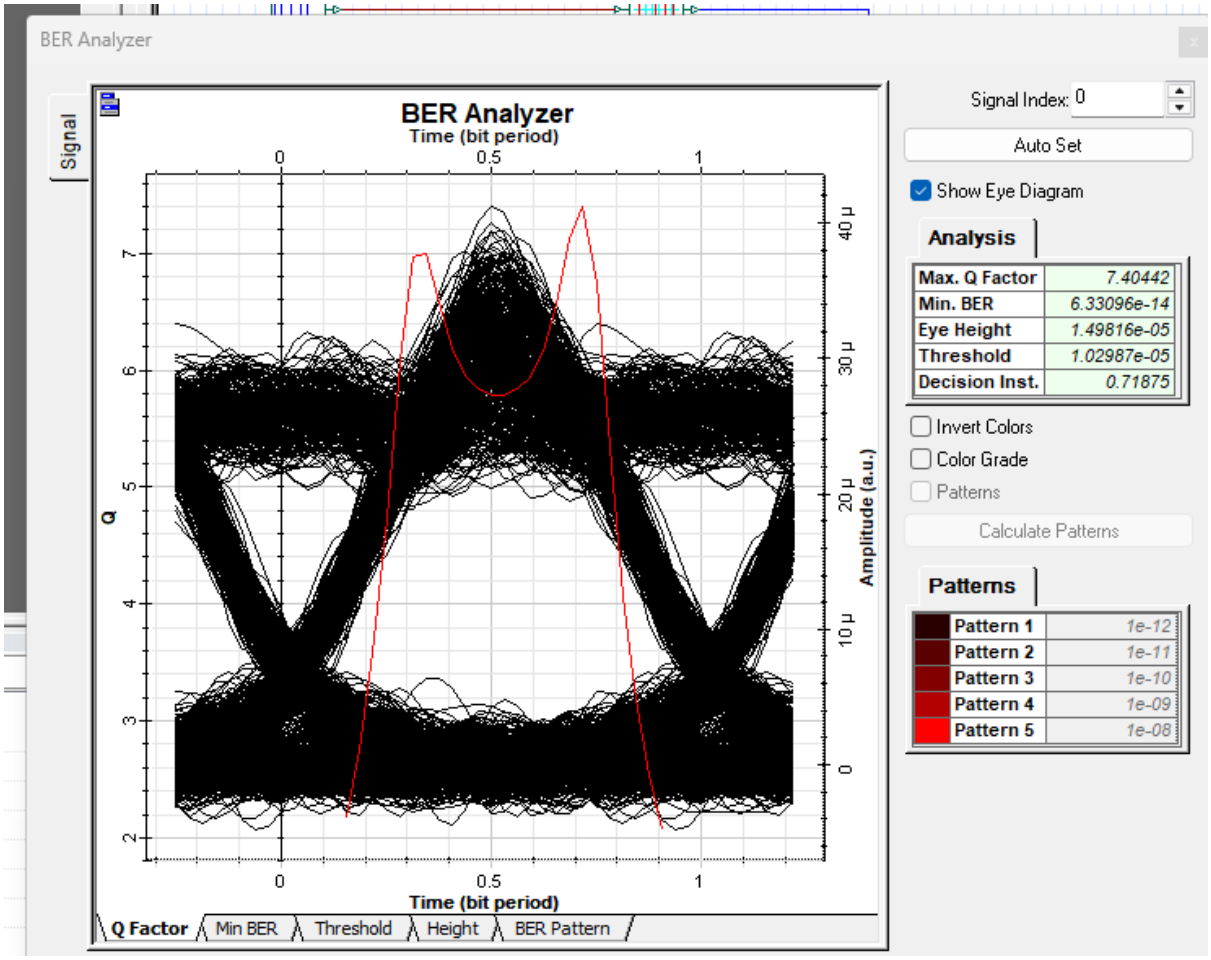
	Min. BER	Max. Q
1	1.47268e-13	7.292

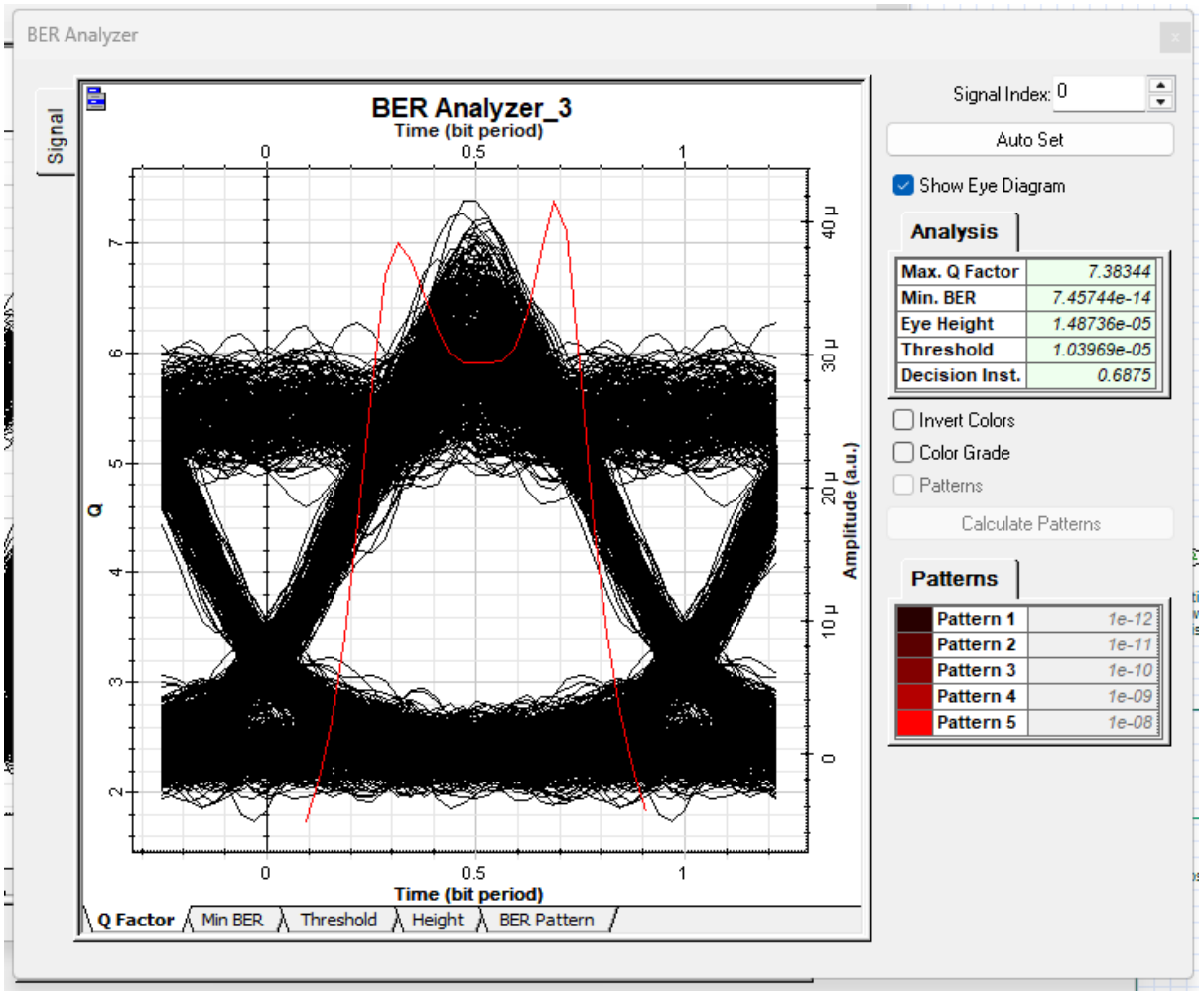
	Min. BER	Max. Q
1	4.58327e-14	7.448

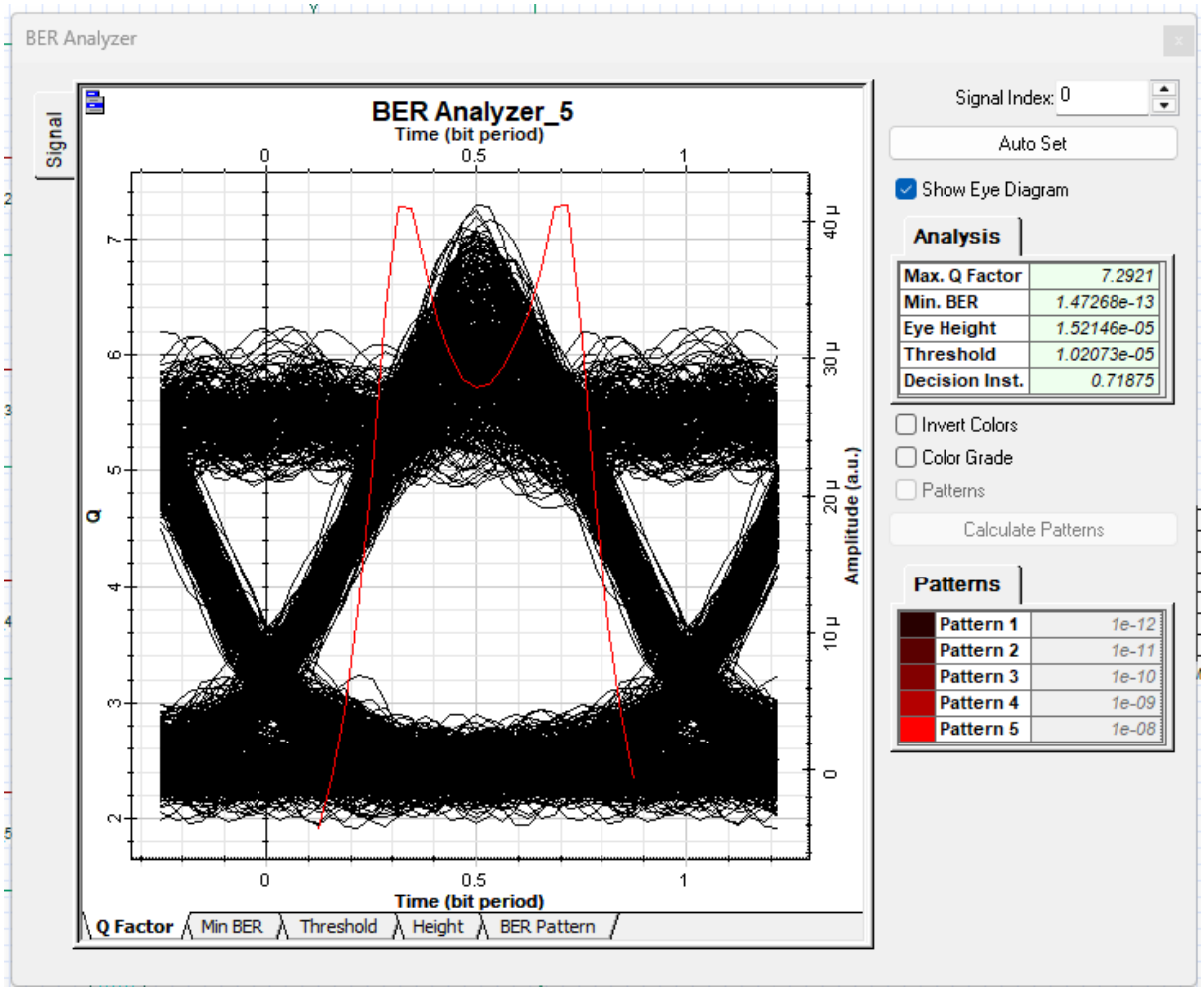
	Min. BER	Max. Q
1	9.57666e-14	7.349

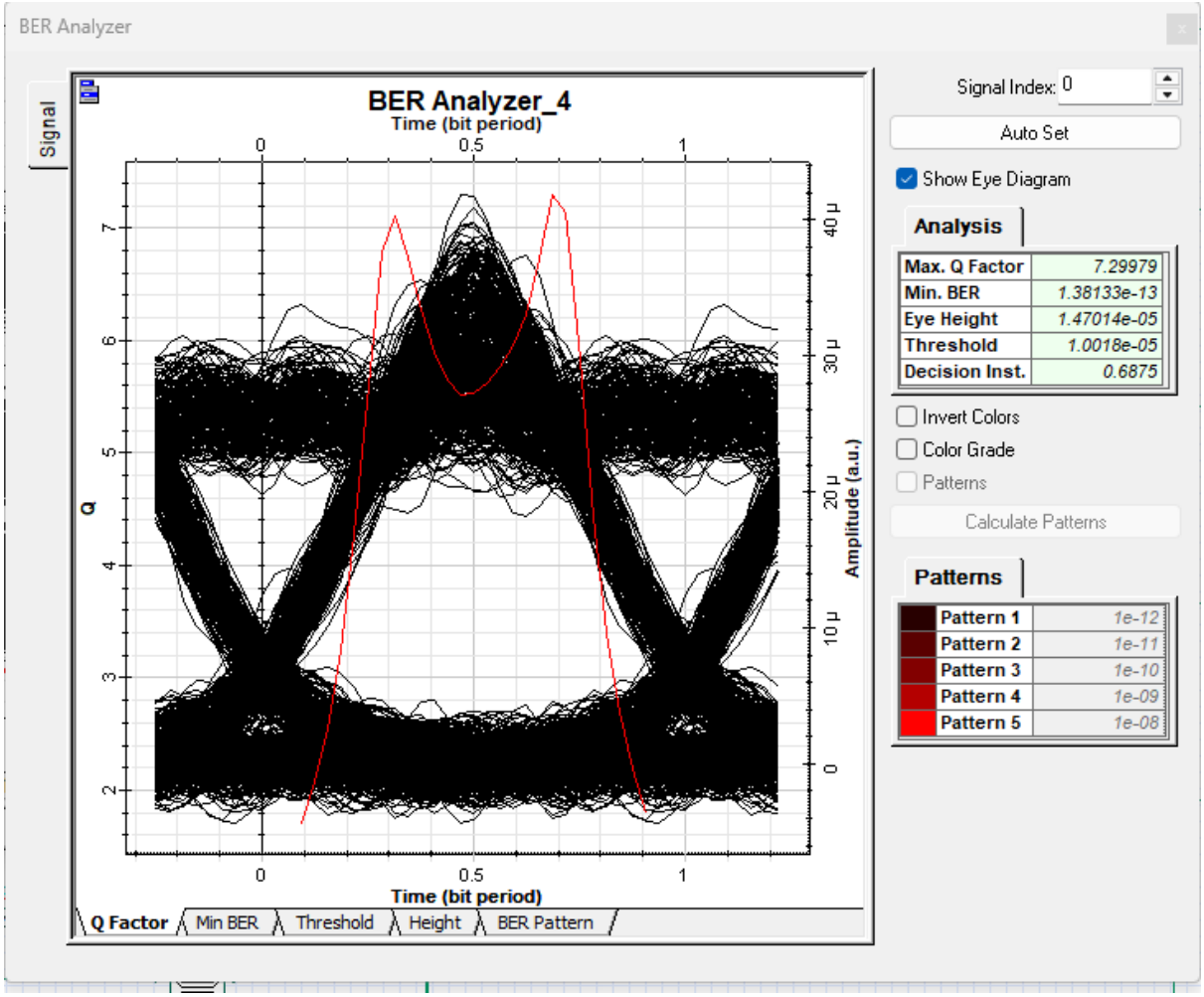


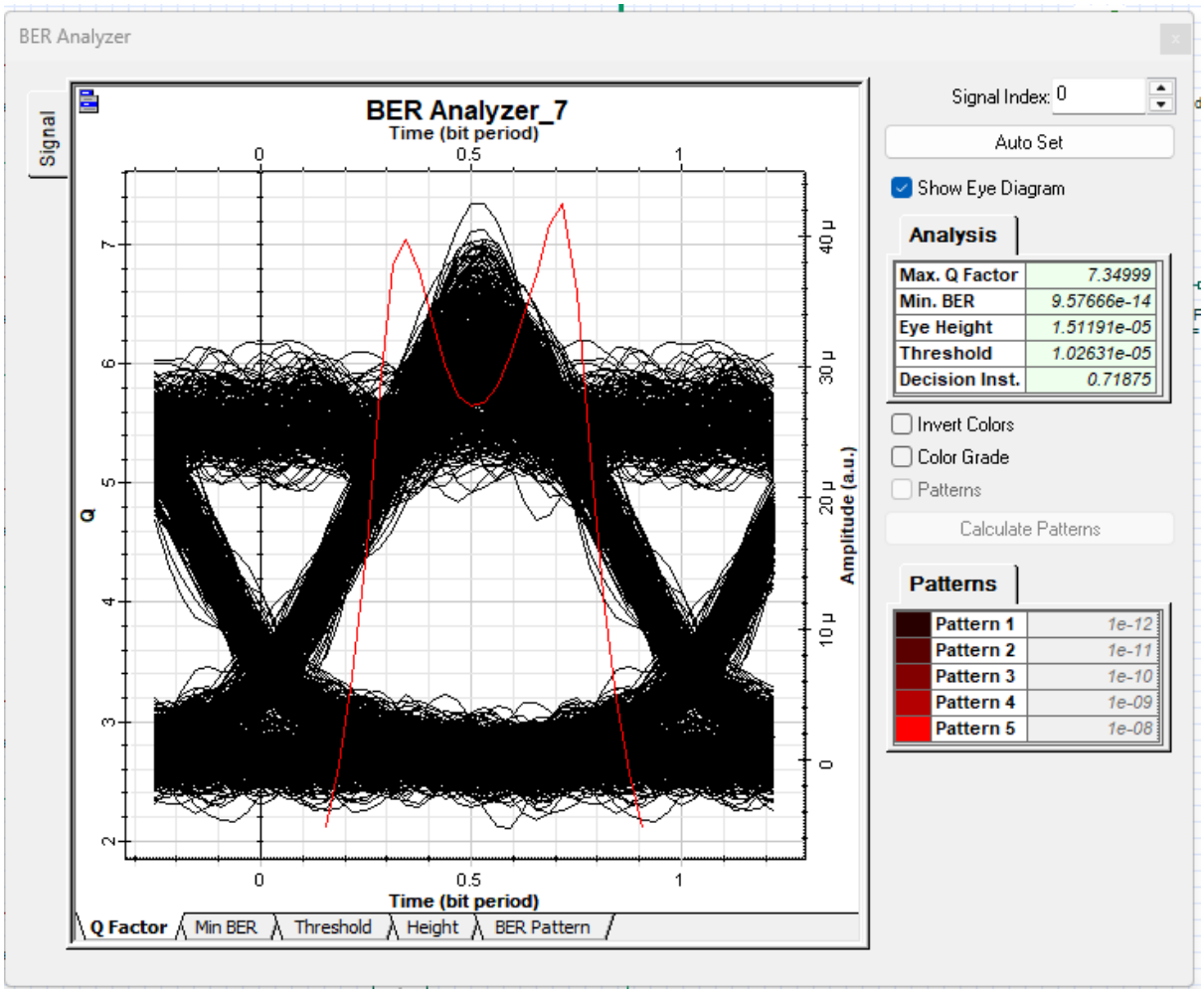

BER Analyzer
Signal
BER Analyzer_2
Time (bit period)
0
0.5
1
Q
7
6
5
4
3
2
40 µ
30 µ
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.19155
Min. BER 3.08506e-13
Eye Height 1.47581e-05
Threshold 1.03422e-05
Decision Inst. 0.3125
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08

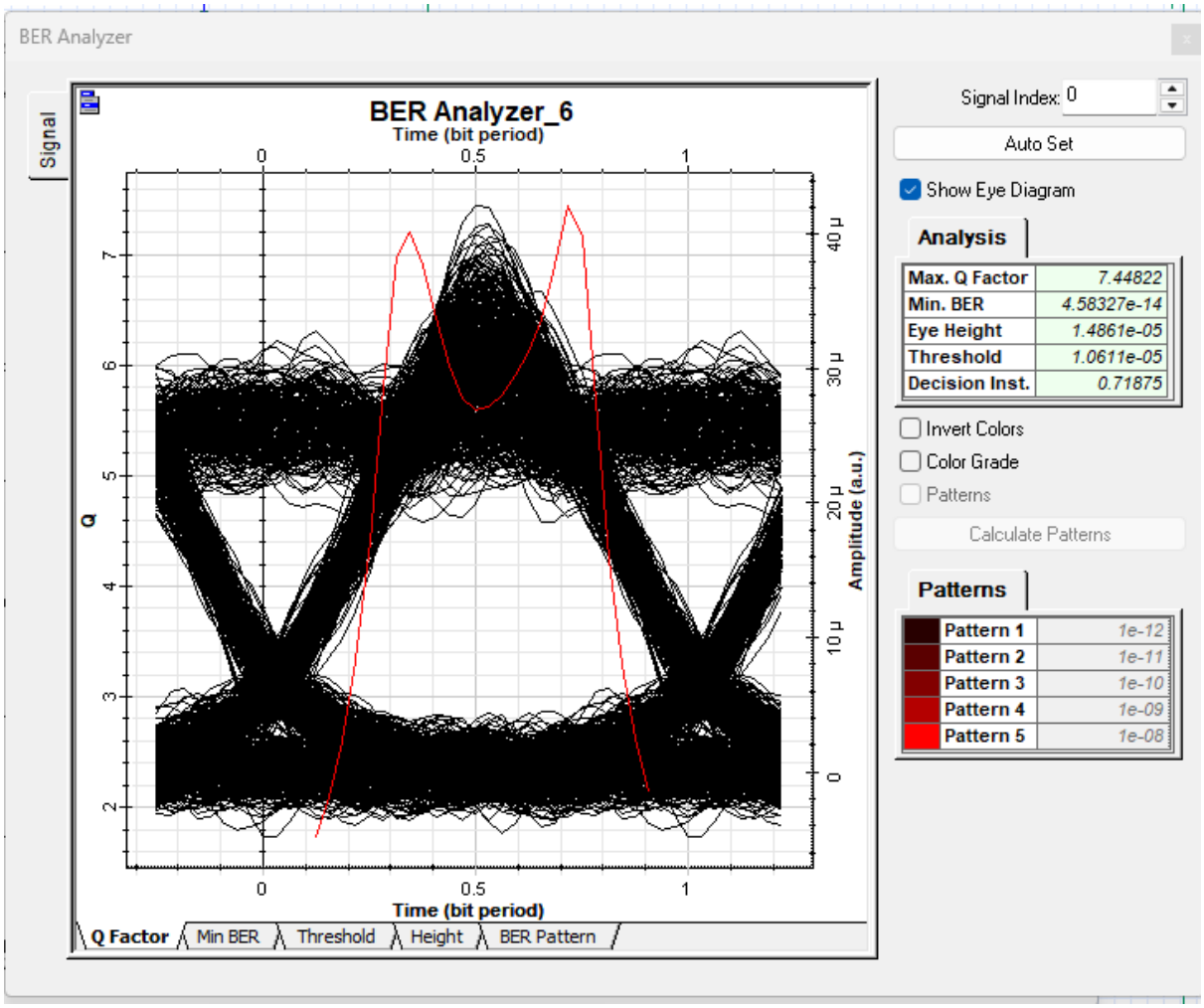

BER Analyzer
Signal
BER Analyzer
Time (bit period)
0
0.5
1
Q
7
6
5
4
3
2
40 µ
30 µ
20 µ
10 µ
0
Amplitude (a.u.)
Time (bit period)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.40442
Min. BER 6.33096e-14
Eye Height 1.49816e-05
Threshold 1.02987e-05
Decision Inst. 0.71875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_3
Time (bit period)
0
0.5
1
Q
7
6
5
4
3
2
40 µ
30 µ
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.38344
Min. BER 7.45744e-14
Eye Height 1.48736e-05
Threshold 1.03969e-05
Decision Inst. 0.6875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_5
Time (bit period)
0
0.5
1
Q
7
6
5
4
3
2
40 µ
30 µ
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.2921
Min. BER 1.47268e-13
Eye Height 1.52146e-05
Threshold 1.02073e-05
Decision Inst. 0.71875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_4
Time (bit period)
0
0.5
1
Q
7
6
5
4
3
2
40 µ
30 µ
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.29979
Min. BER 1.38133e-13
Eye Height 1.47014e-05
Threshold 1.0018e-05
Decision Inst. 0.6875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_7
Time (bit period)
0
0.5
1
Q
7
6
5
4
3
2
40 µ
30 µ
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.34999
Min. BER 9.57666e-14
Eye Height 1.51191e-05
Threshold 1.02631e-05
Decision Inst. 0.71875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08



CWDM:

Bit Rate- 2.5 Gbps

Line Encoding: NRZ

Modulation: External Modulation

Channel Spacing= 20 nm

λ_1= 1550 nm

λ_2= λ_1 nm + 20 nm=

λ_3= λ_2 nm + 20 nm

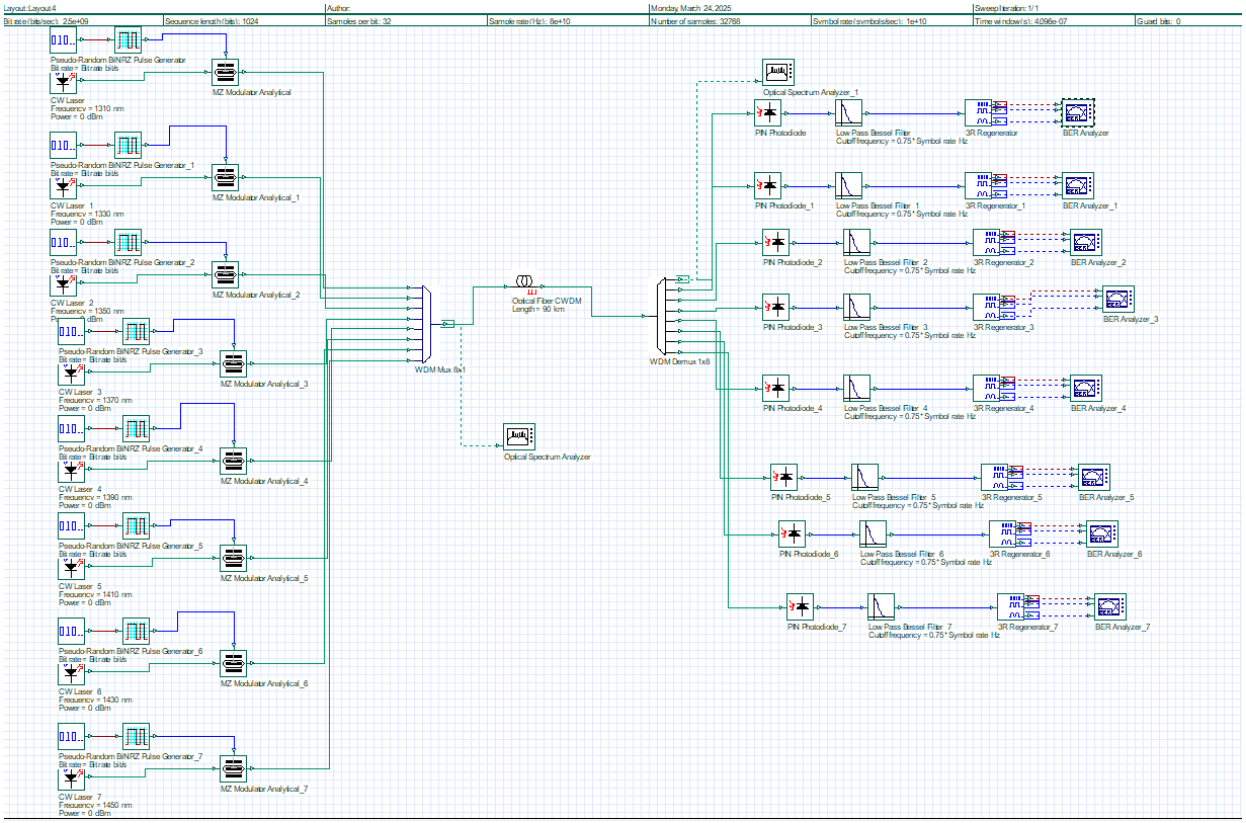
λ_4= $\lambda_3$nm + 20 nm

λ_5= $\lambda_4$nm + 20 nm

λ_6= λ_5 nm + 20 nm

λ_7= $\lambda_6$nm + 20 nm

$\lambda_8 = \lambda_7$ nm + 20 nm



	Min. BER	Max. Q
1	2.48096e-11	6.57183

	Min. BER	Max. Q
1	1.49441e-10	6.29797

	Min. BER	Max. Q
1	6.4015e-11	6.42861
2		

	Min. BER	Max. Q
1	2.11668e-10	6.2447
2		

	Min. BER	Max. Q
1	1.5255e-10	6.2951
2		

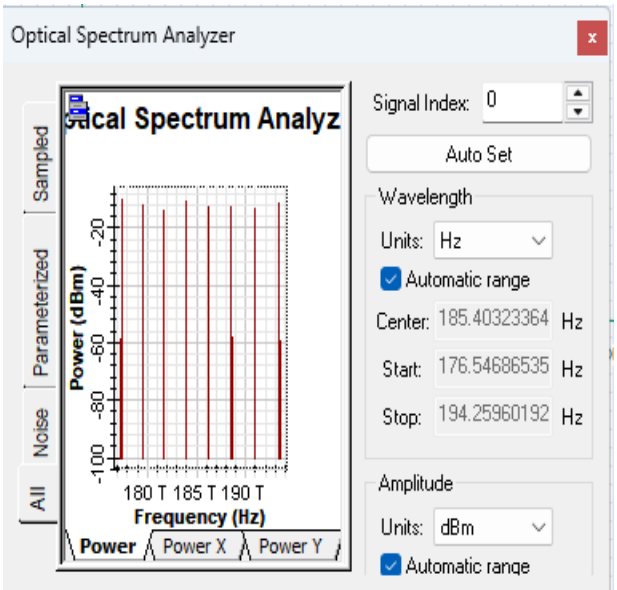
	Min. BER	Max. Q
1	2.64783e-10	6.20866
2		

	Min. BER	Max. Q
1	8.32495e-11	6.3883
2		

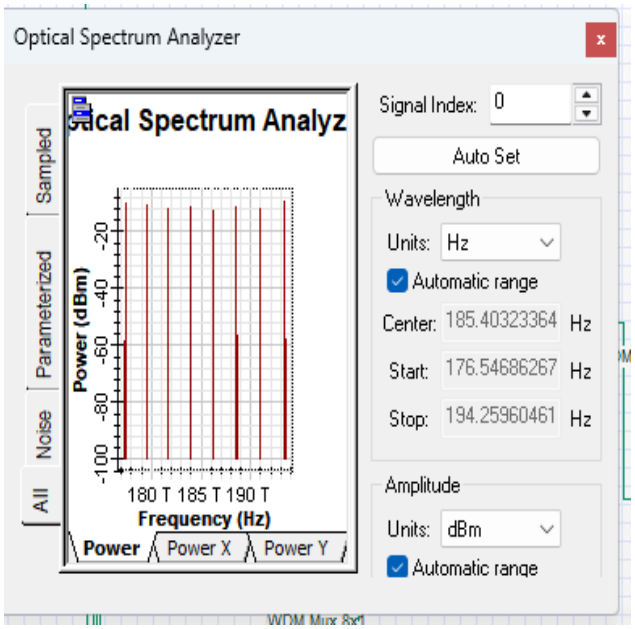
	Min. BER	Max. Q
1	7.72779e-11	6.400
2		

Table:CWDM

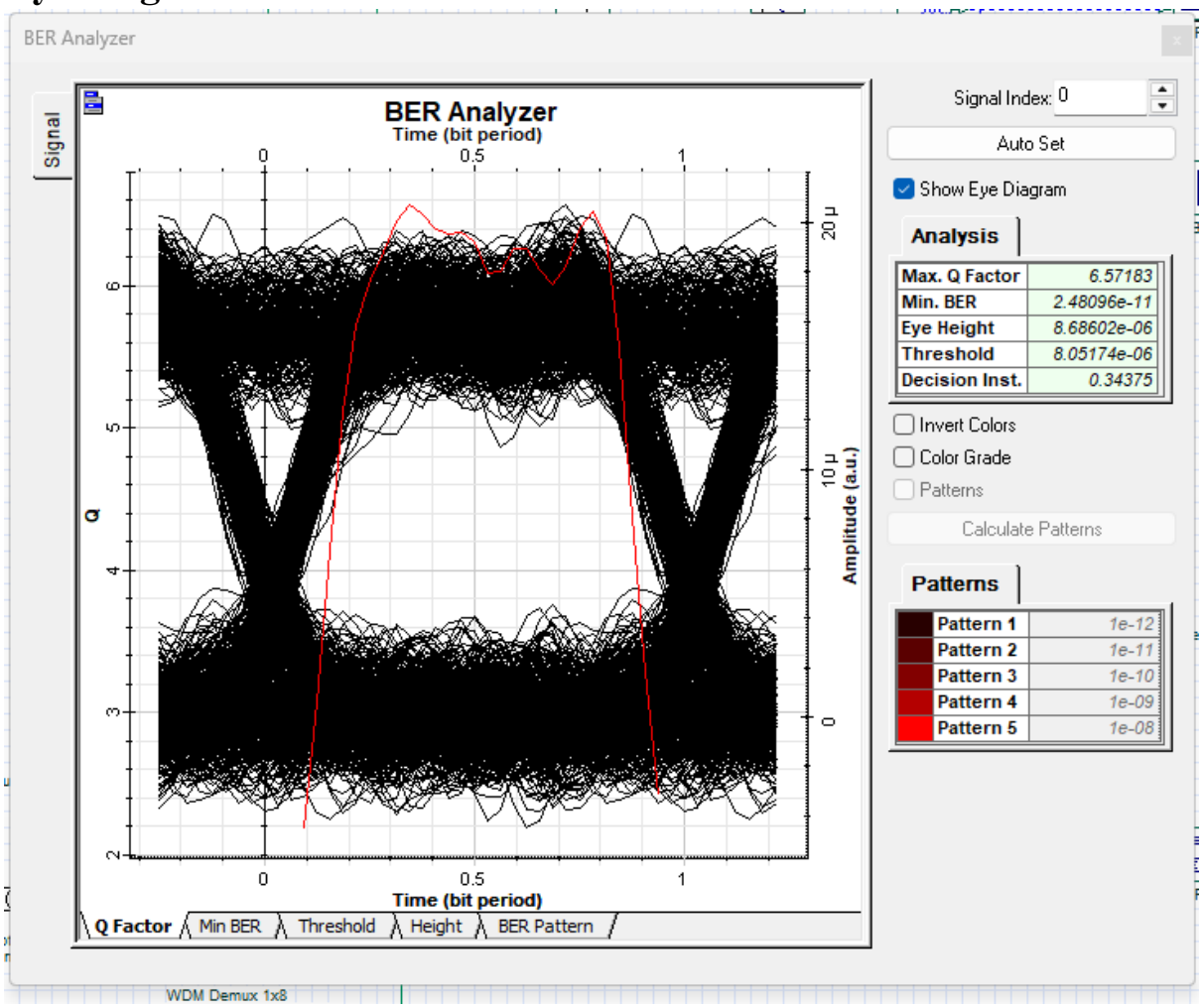
SPECTRUM: CWDM

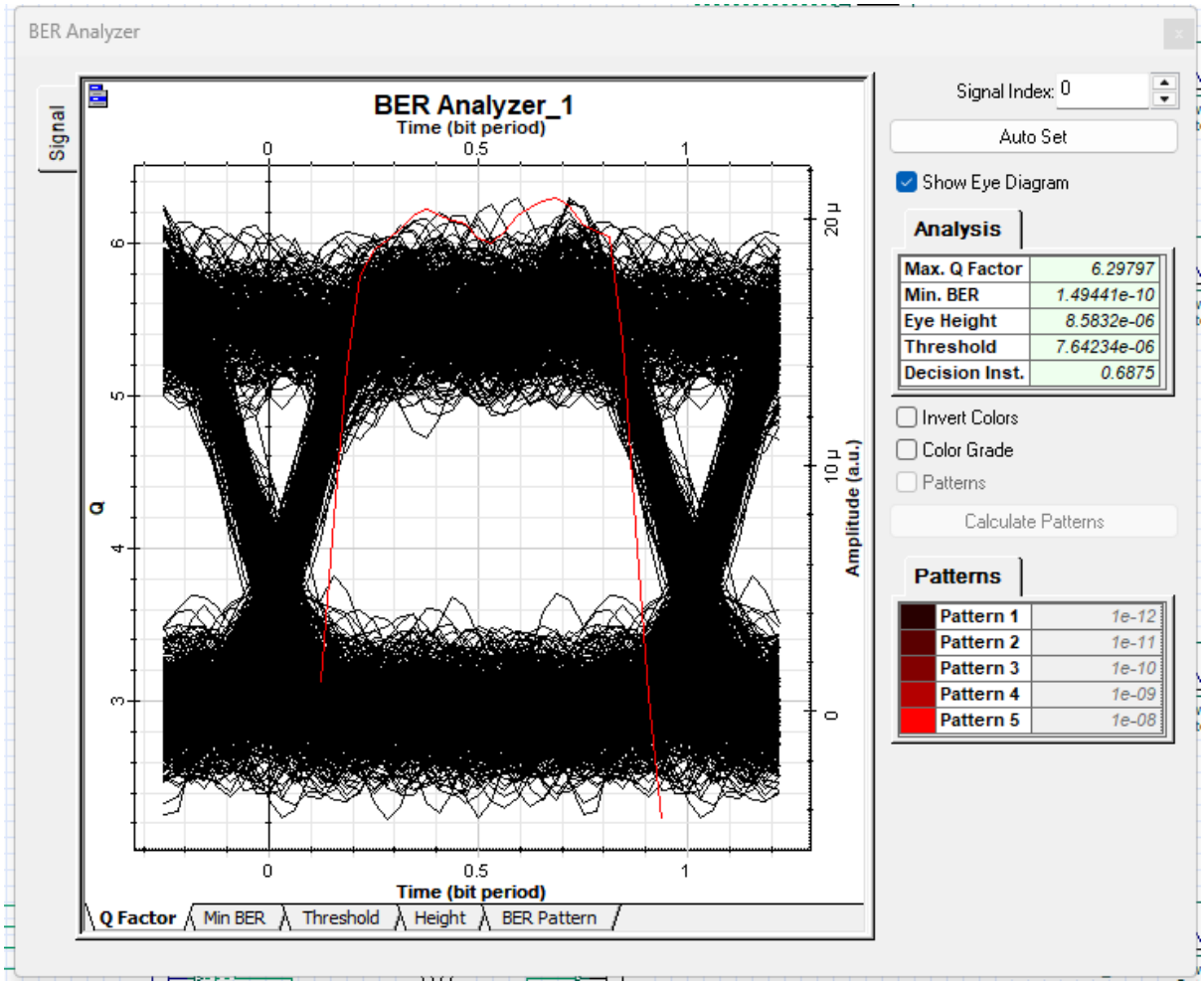


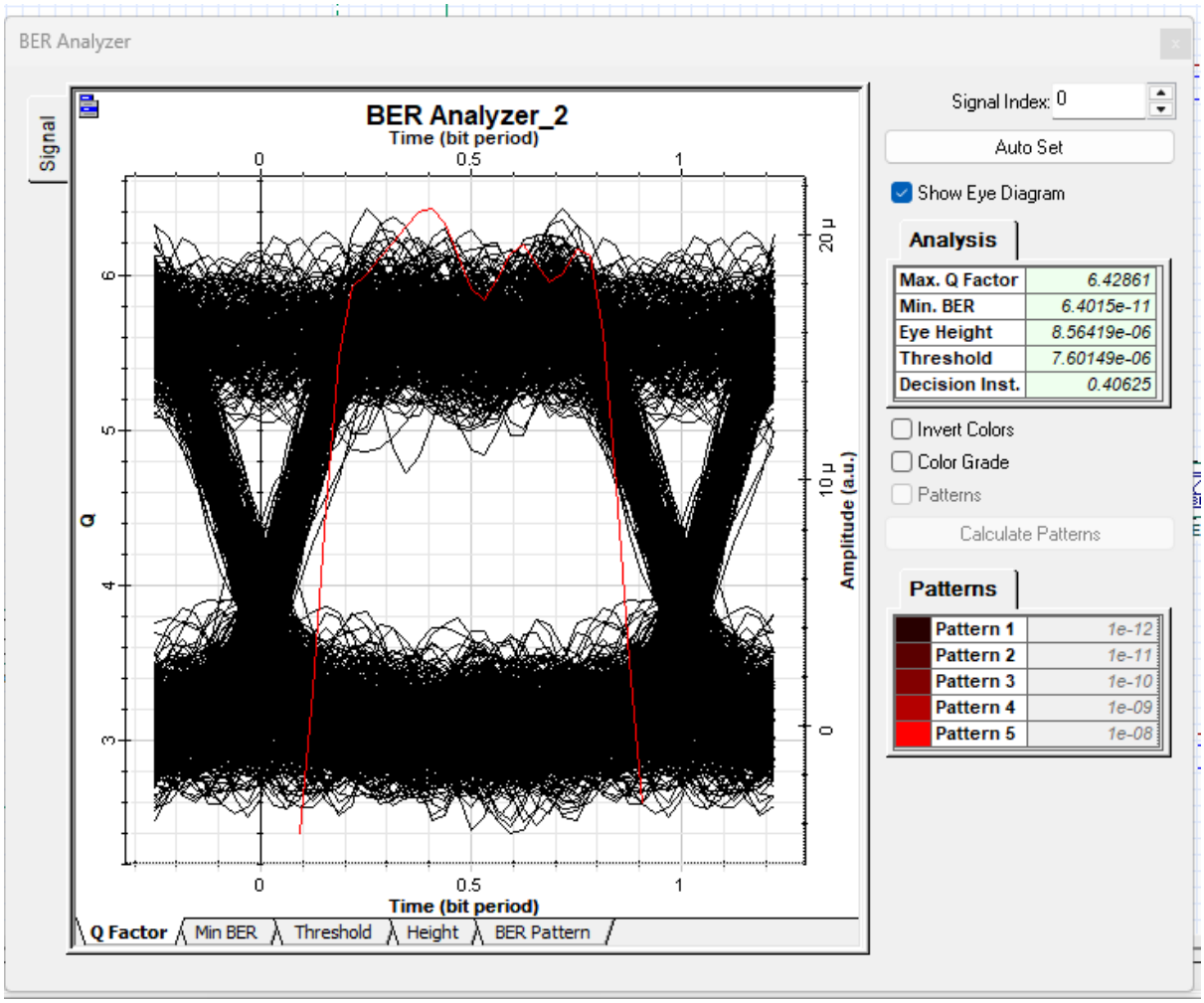
CWDM Channel Spacing < User Bandwidth

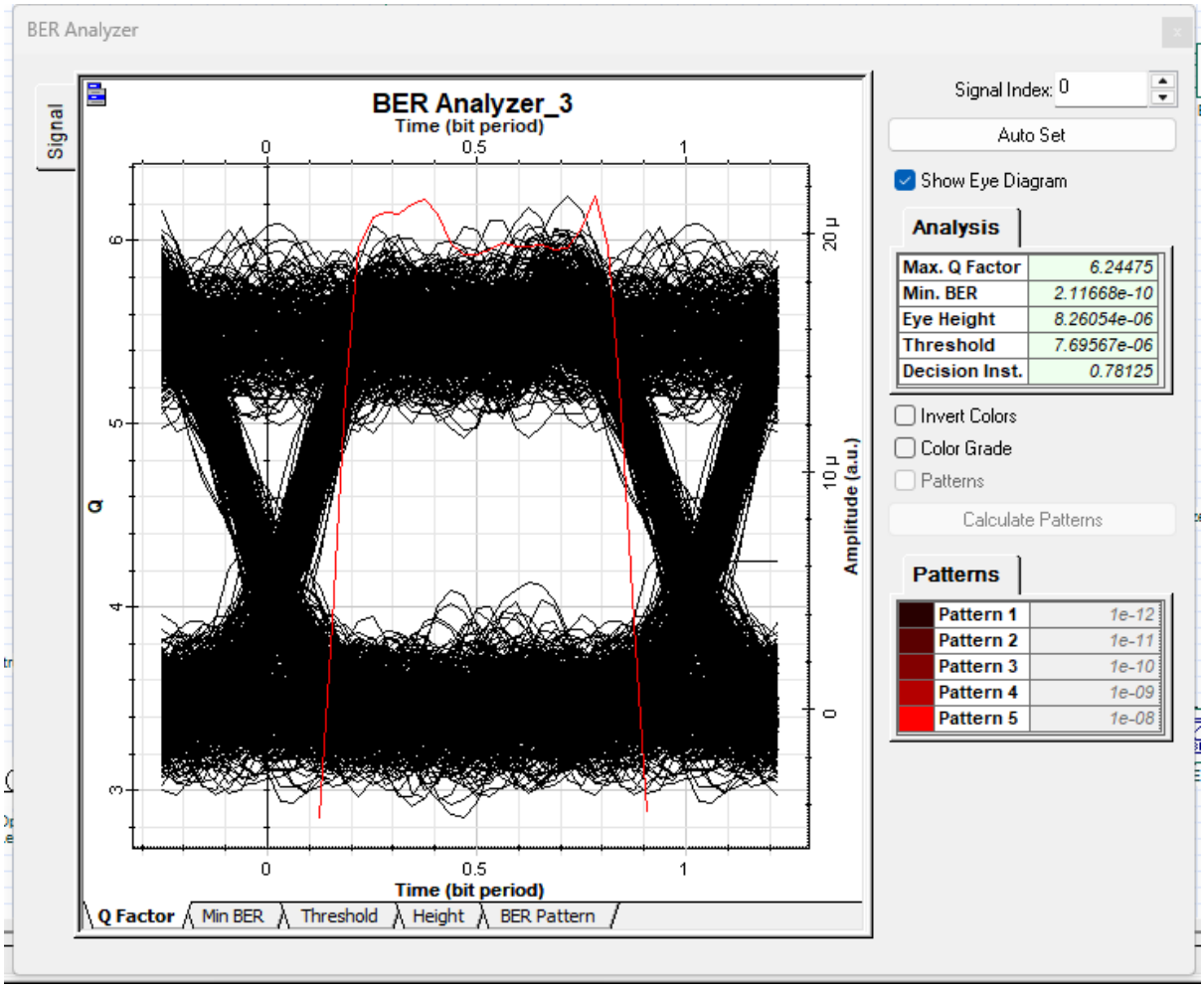


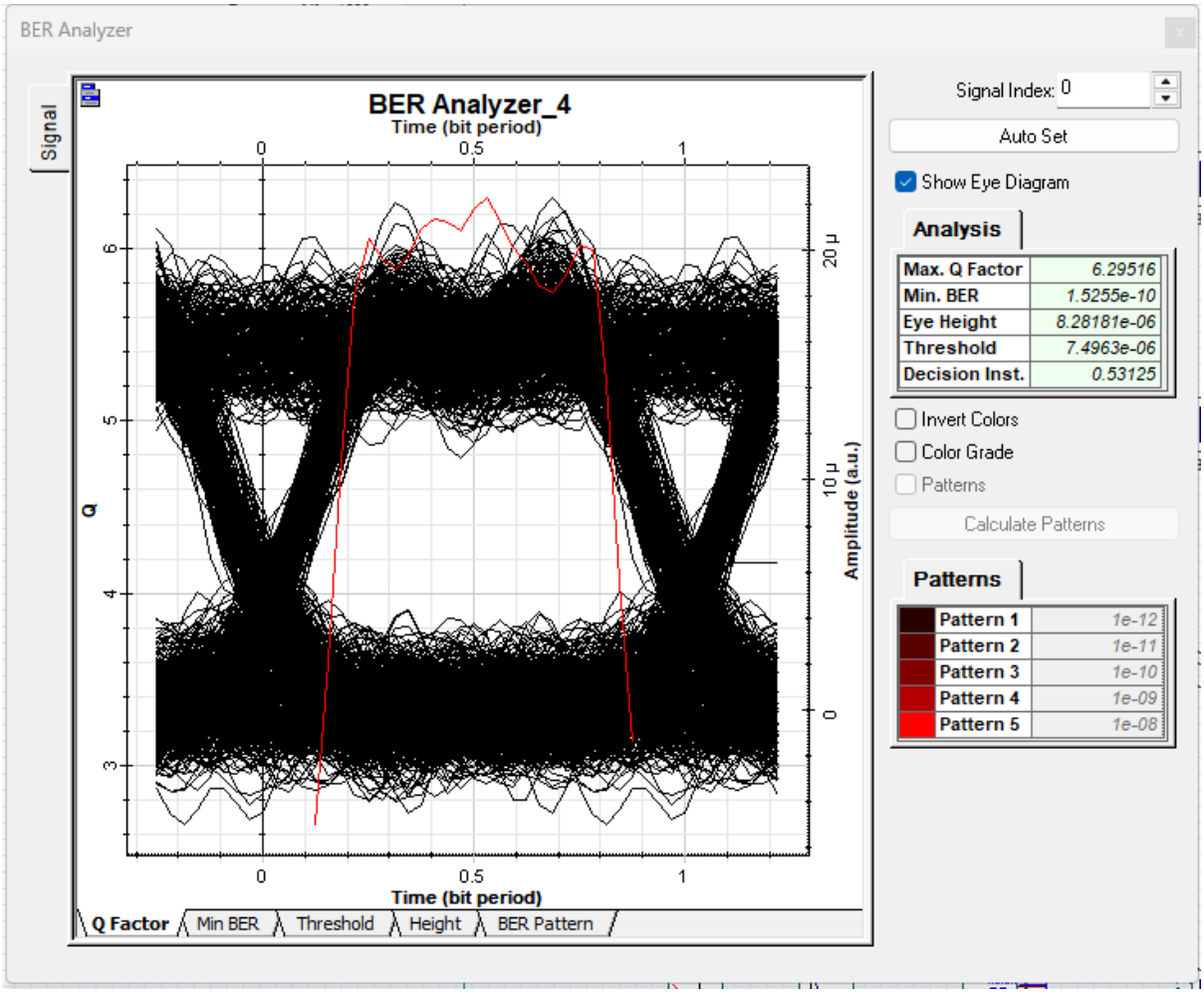
Eye diagram:

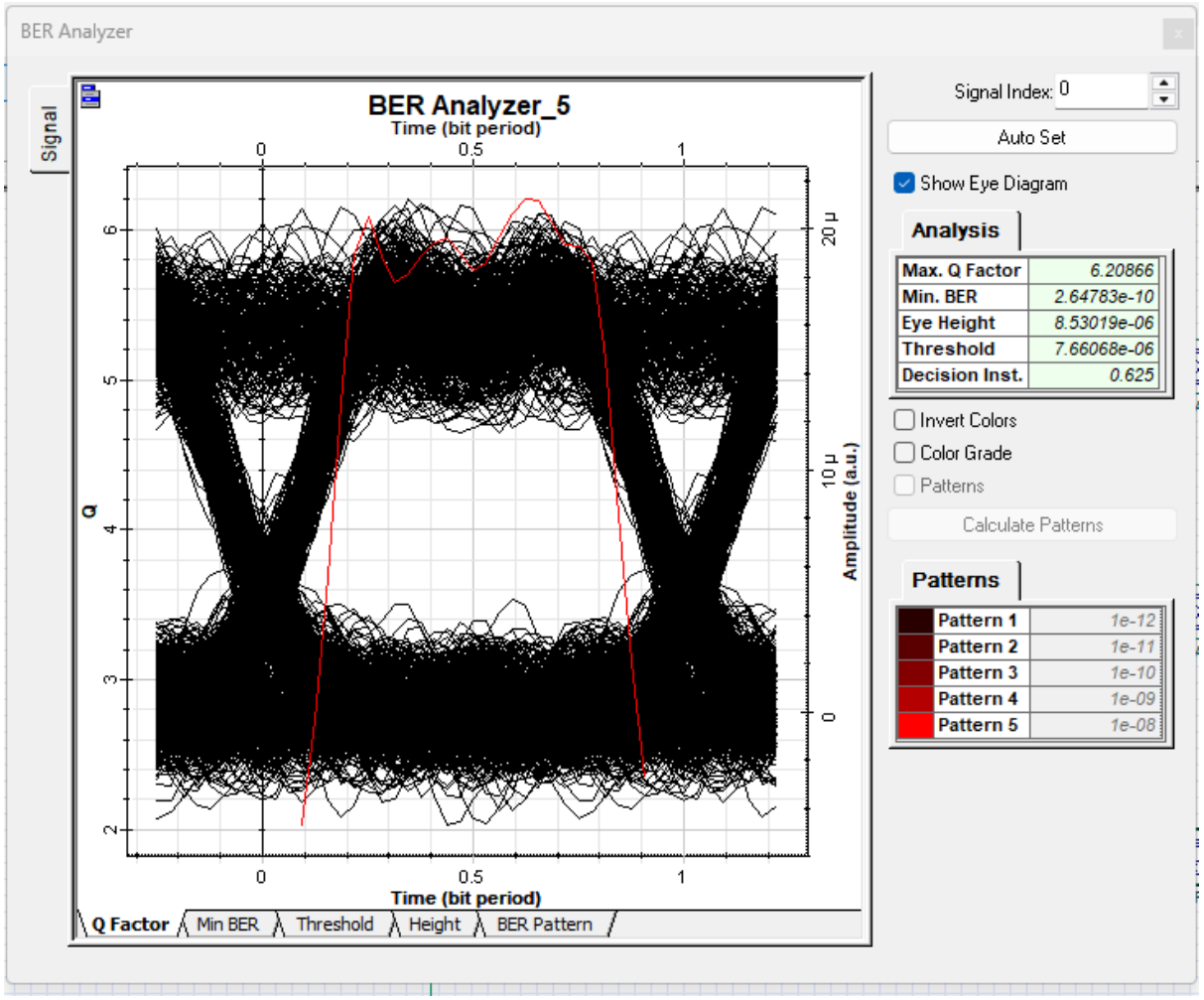

BER Analyzer
Signal
BER Analyzer
Time (bit period)
0
0.5
1
Q
6
5
4
3
2
20 µ
10 µ
0
Amplitude (a.u.)
Time (bit period)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.57183
Min. BER 2.48096e-11
Eye Height 8.68602e-06
Threshold 8.05174e-06
Decision Inst. 0.34375
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08
WDM Demux 1x8

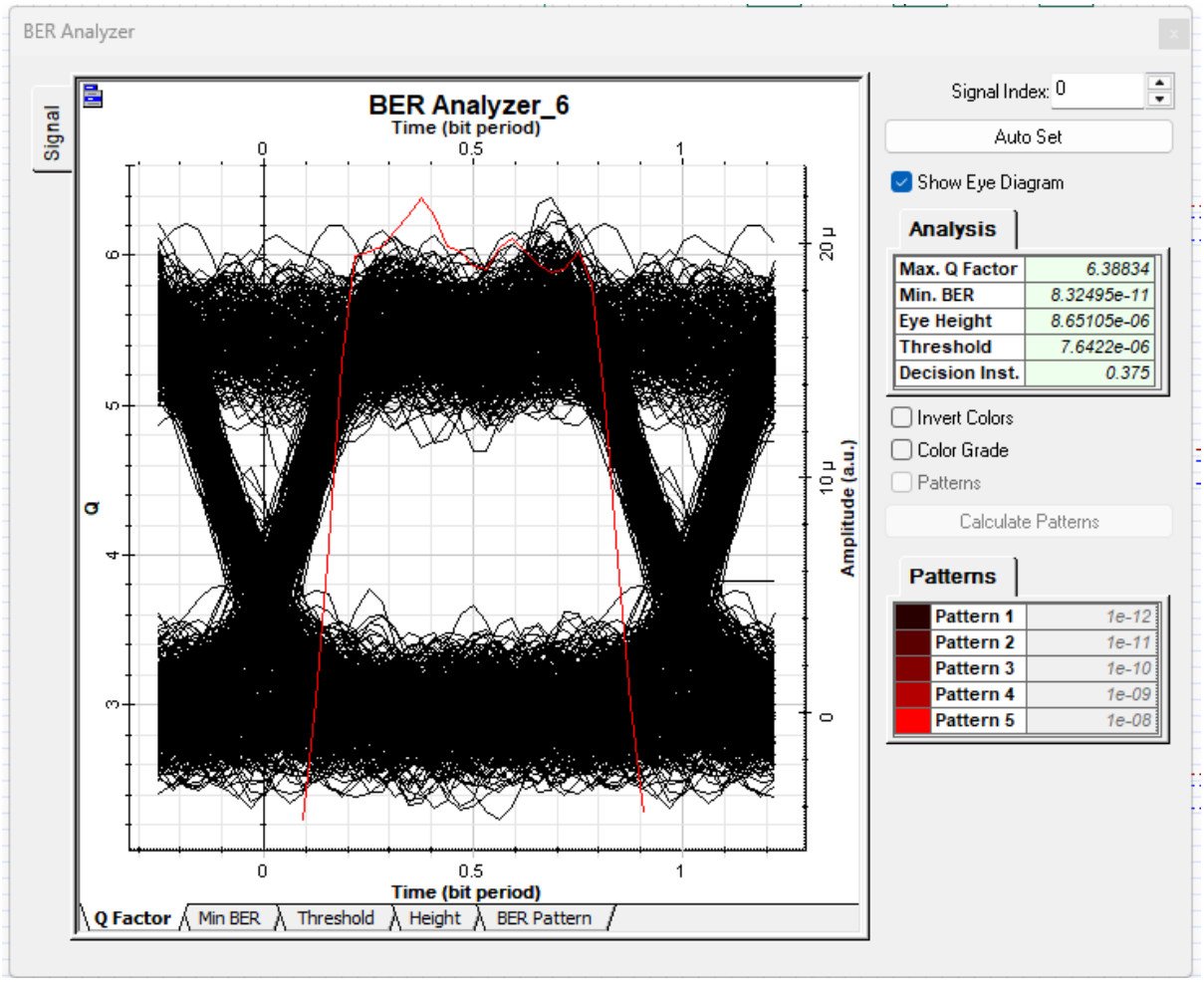

BER Analyzer
Signal
BER Analyzer_1
Time (bit period)
0
0.5
1
Q
6
5
4
3
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.29797
Min. BER 1.49441e-10
Eye Height 8.5832e-06
Threshold 7.64234e-06
Decision Inst. 0.6875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08

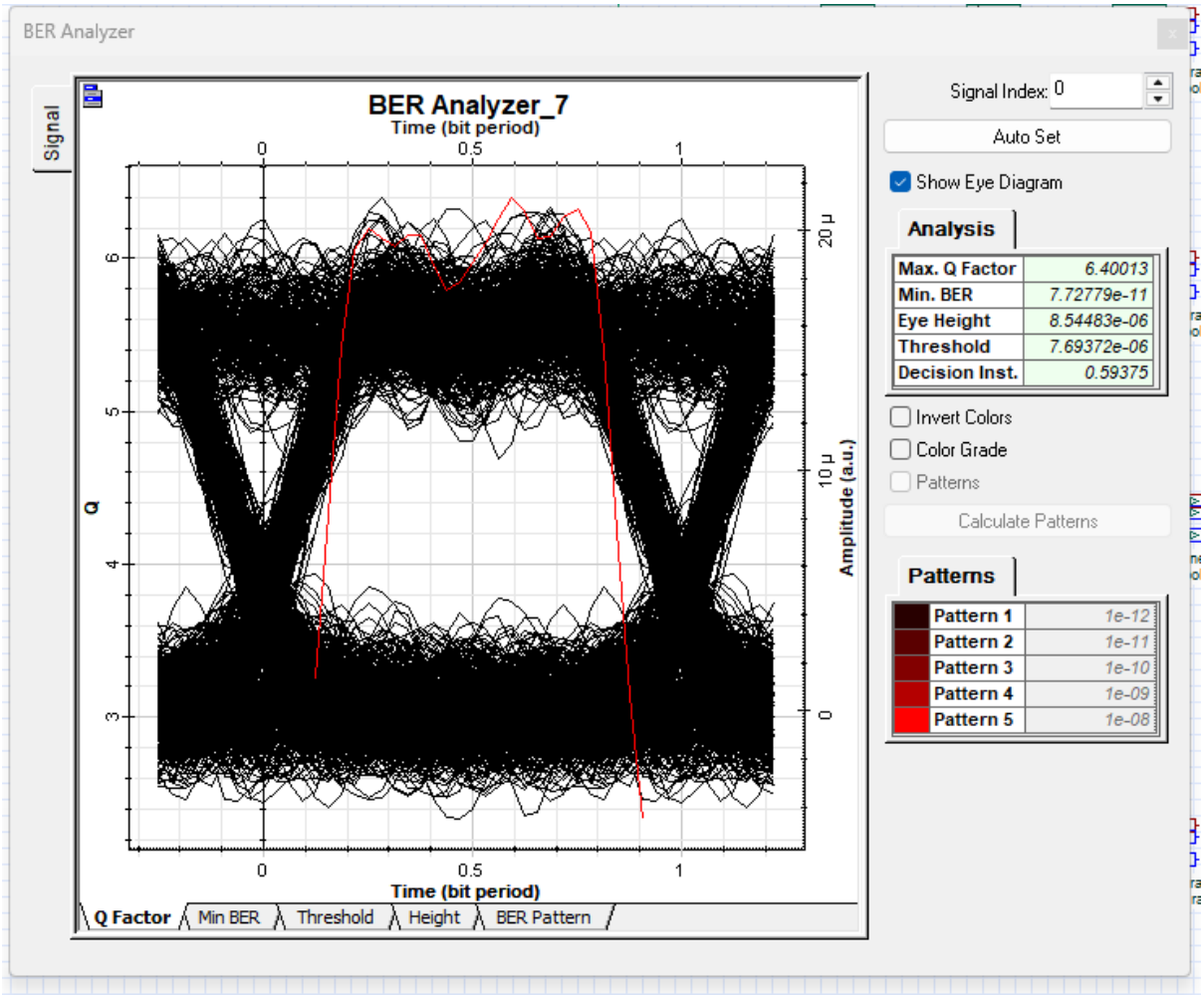

BER Analyzer
Signal
BER Analyzer_2
Time (bit period)
0
0.5
1
Q
6
5
4
3
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.42861
Min. BER 6.4015e-11
Eye Height 8.56419e-06
Threshold 7.60149e-06
Decision Inst. 0.40625
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_3
Time (bit period)
0
0.5
1
Q
6
5
4
3
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.24475
Min. BER 2.11668e-10
Eye Height 8.26054e-06
Threshold 7.69567e-06
Decision Inst. 0.78125
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_4
Time (bit period)
0
0.5
1
Q
6
5
4
3
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.29516
Min. BER 1.5255e-10
Eye Height 8.28181e-06
Threshold 7.4963e-06
Decision Inst. 0.53125
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_5
Time (bit period)
0
0.5
1
Q
6
5
4
3
2
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.20866
Min. BER 2.64783e-10
Eye Height 8.53019e-06
Threshold 7.66068e-06
Decision Inst. 0.625
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_6
Time (bit period)
0
0.5
1
Q
6
5
4
3
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.38834
Min. BER 8.32495e-11
Eye Height 8.65105e-06
Threshold 7.6422e-06
Decision Inst. 0.375
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08


BER Analyzer
Signal
BER Analyzer_7
Time (bit period)
0
0.5
1
Q
6
5
4
3
20 µ
10 µ
0
Amplitude (a.u.)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 6.40013
Min. BER 7.72779e-11
Eye Height 8.54483e-06
Threshold 7.69372e-06
Decision Inst. 0.59375
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-12
Pattern 2 1e-11
Pattern 3 1e-10
Pattern 4 1e-09
Pattern 5 1e-08

Experiment No: 6

DESIGN ANALYSIS OF DISPERSION COMPENSATION

Design & simulate the Optical Communication Link with dispersion Compensation fiber and Analyse the performance at different Configuration (Pre, Post, Symmetric dispersion Compensation)

Experimentation:

1. Plot the spectrum of the transmitter and receiver comment on

the result.

2. Note down the BER of system at

Different Configuration**: Pre, Post, Symmetric dispersion Compensation**

Operating WaveLength : **1550nm**

Data rate:2.5Gbps, 10Gbps

3. Write your inference on the result obtained.

Aim :

To study performance of optical fiber link before and after adding dispersion compensation fibers of all types (Pre, Post, Hybrid) and perform comparative analysis in Optisystem software

Software/Components Required:

Optisystem 22.0Verasion

Theory:

The pulse broadening effect of chromatic dispersion causes the signals in the adjacent bit periods to overlap. This is called intersymbol interference (ISI).Broadening is a function of distance as well as dispersion parameter D. The basic idea for DCF: the positive dispersion in a conventional fiber (say ~ 17 ps/(km-nm) in the 1550 nm window) can be compensated for by inserting a fiber with negative dispersion.

The dispersion parameter is given in ps/nm/km and changes from fiber to fiber. It is also a function of wavelength. D is usually about 16.75 ps/nm/km in the 1.55 µm wavelength range for a standard single mode fiber (SMF).The three different schemes, pre-, post-, and symmetrical compensation are used to compensate the fiber dispersion

Optical amplifiers are used after each fiber to compensate for the span loss. The dispersion parameter of SMF is 120 km long and 16.75 ps/nm-km. Therefore, total accumulated dispersion is 16x120 = 1920 ps/nm. This much dispersion can be compensated by using a 24 km long DCF with -80 ps/km-nm dispersion. Total transmission distance is 120x2 =240 km for each cases. In the post-compensation case, DCF is placed after SMF. In the symmetrical compensation case, fiber placement follows the sequence of SMF,DCF, DCF, SMF.

Design:

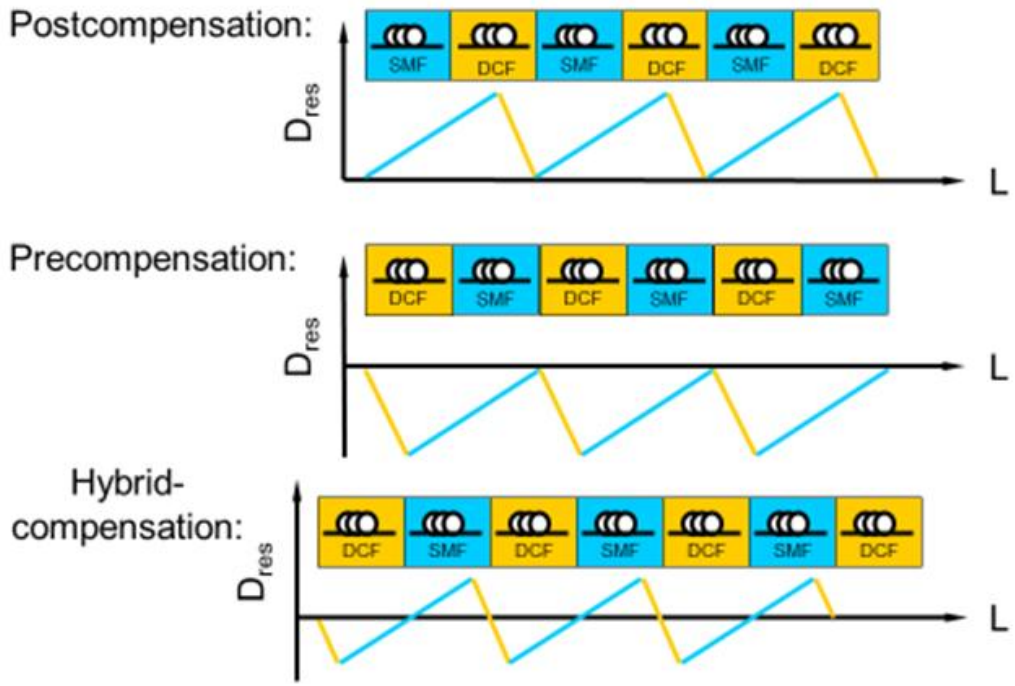
Dispersion Compensation:

$D_0L_0+D_1L_1=0$

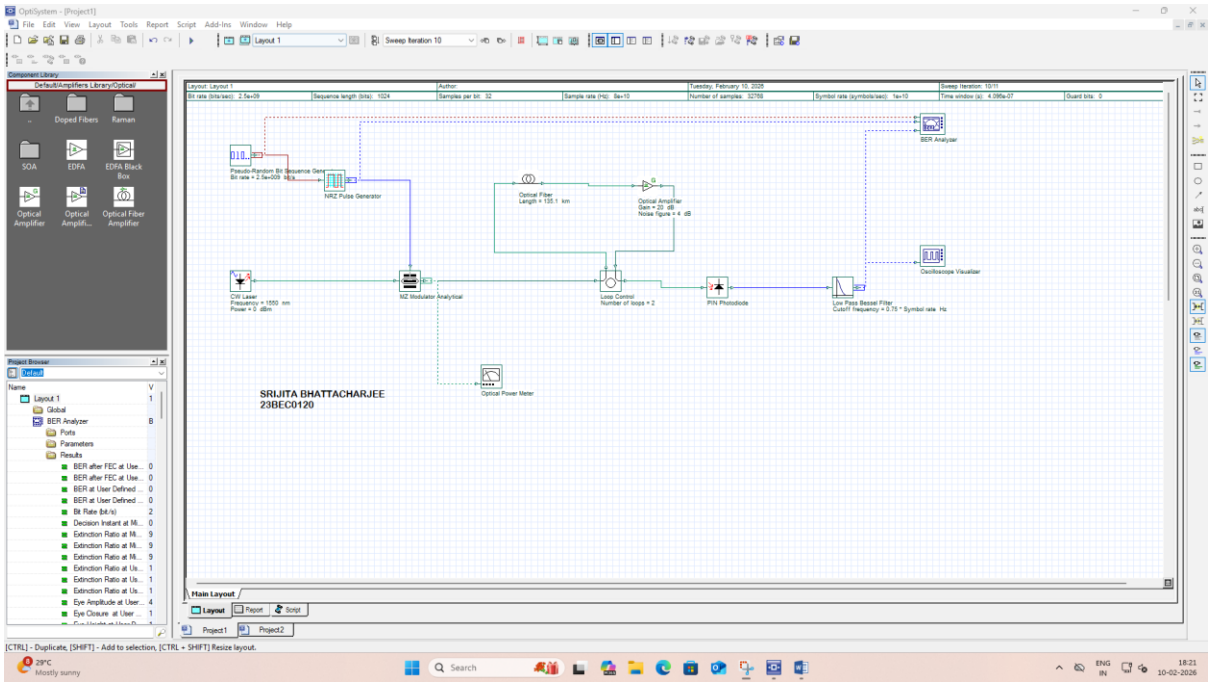
D_0 is the dispersion (16.75 ps/nm-km) and L_0 is the length of the SMF (120 Km).

D_1 is the dispersion (-80 ps/km-nm) and L_1 is the length of the DCF (24 Km)

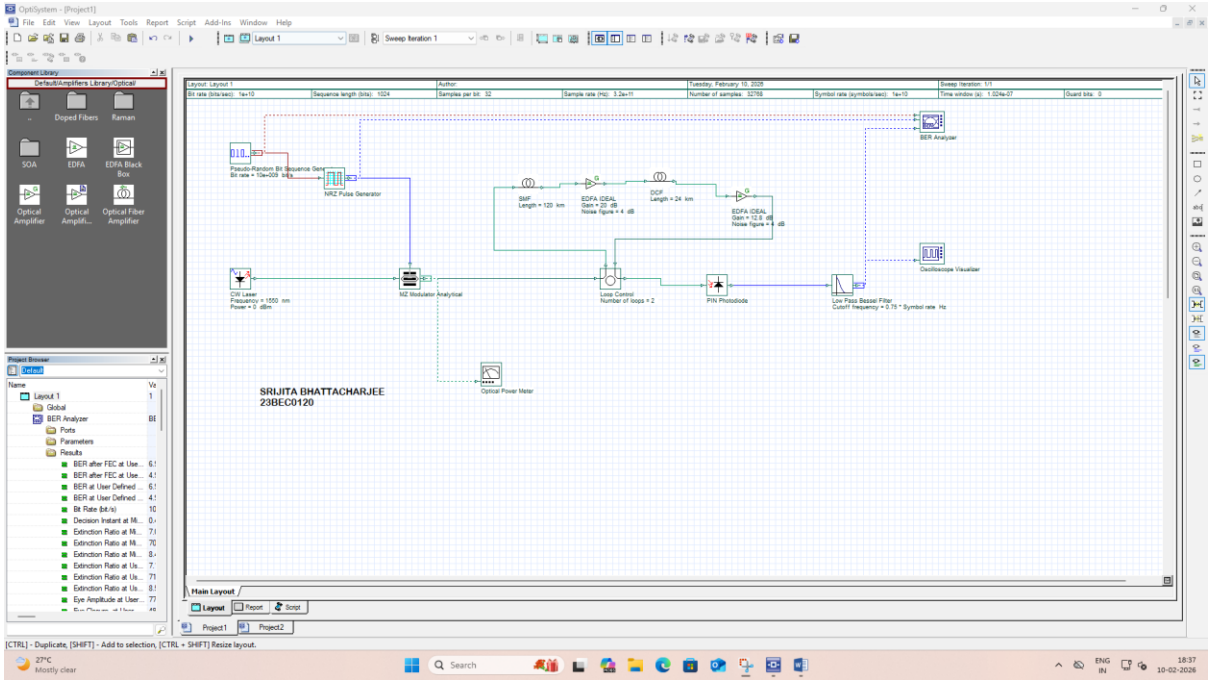
Block Diagram:



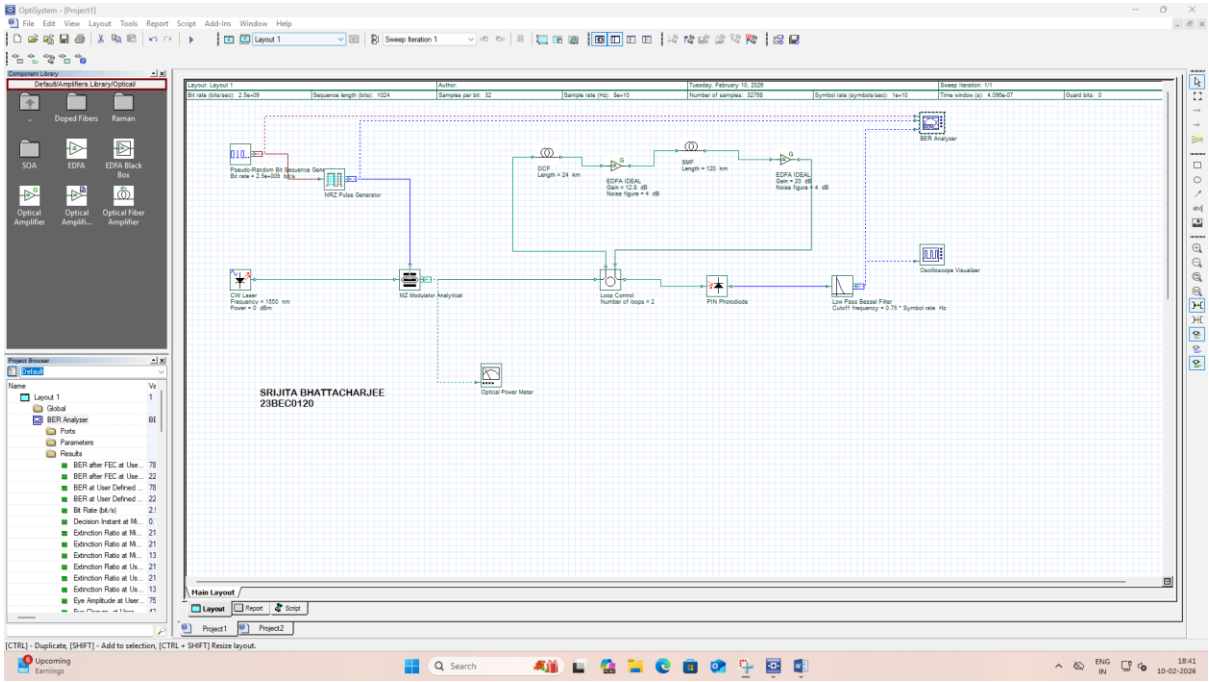
BeforeCompensation



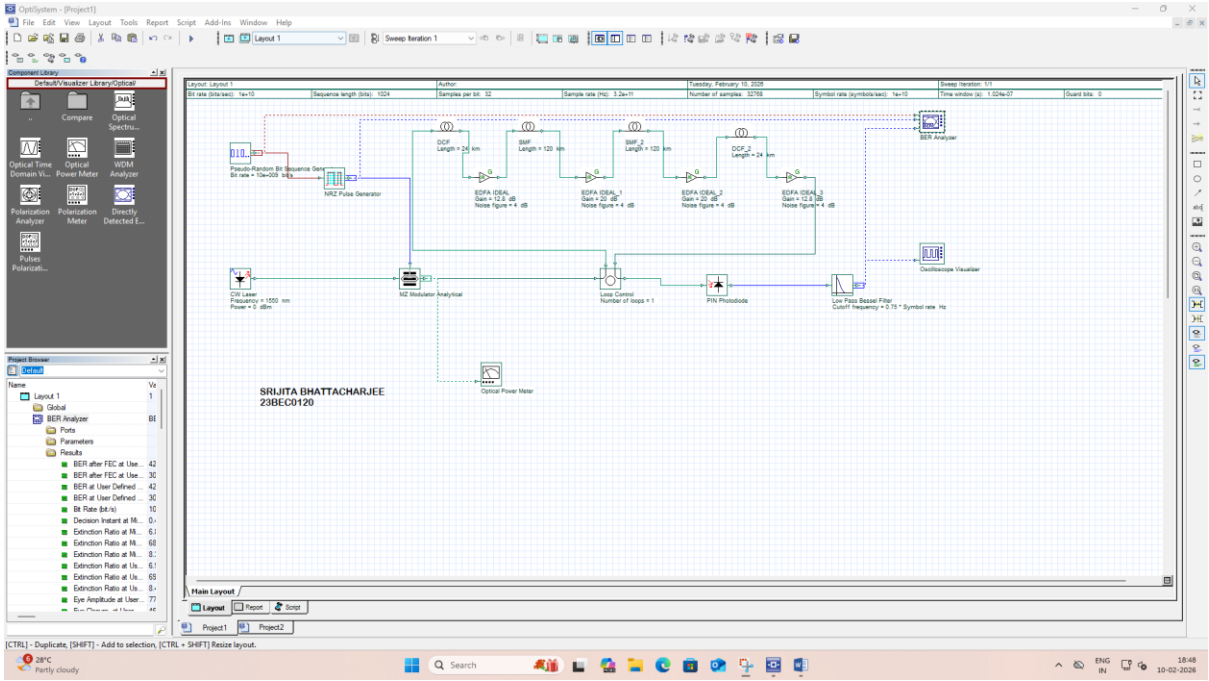
Dispersion post-compensation



Dispersion Pre-compensation



Dispersion Hybrid-compensation



Observation:

Data Rate: 2.5 Gbps

Compensation	BER(x10^-22)	Q factor
Without compensation	3.6407	4.913
Pre Compensation	2.128	8.643
Post Compensation	8.599	9.497
Hybrid Compensation	2.098	7.829

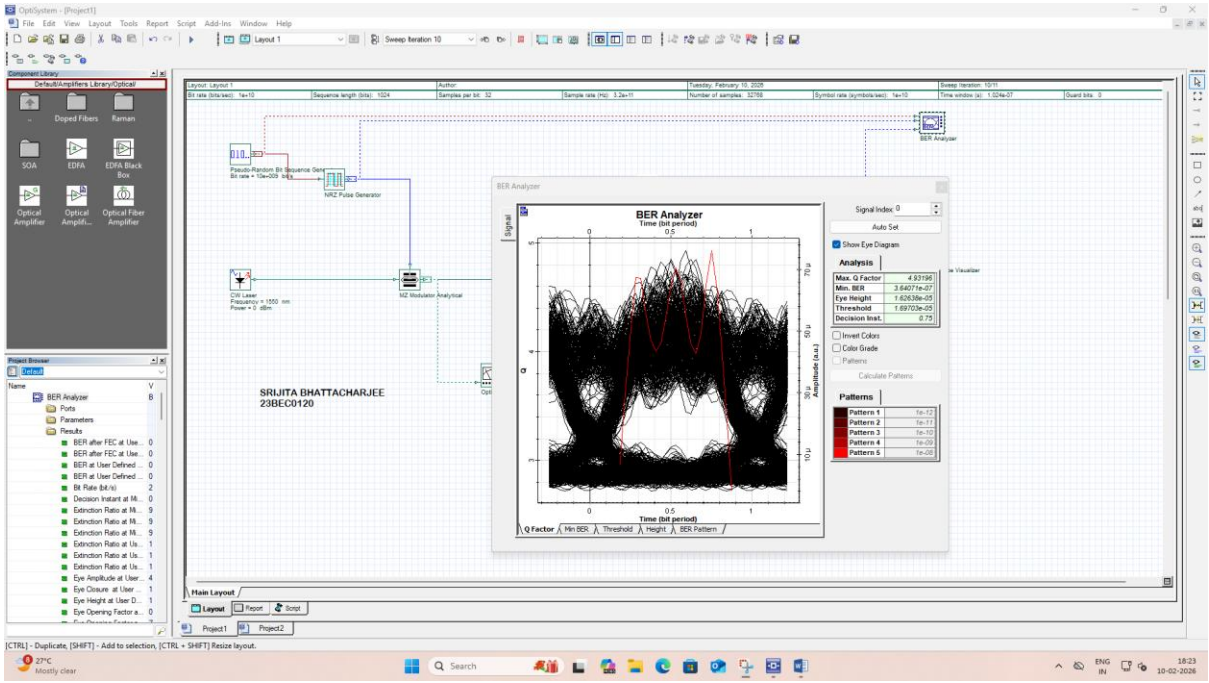
Data Rate: 10 Gbps

Compensation	BER(x10^-16)	Q factor
Without compensation	1	0
Pre Compensation	2.208	7.22
Post Compensation	2.604	8.109
Hybrid Compensation	2.098	7.826

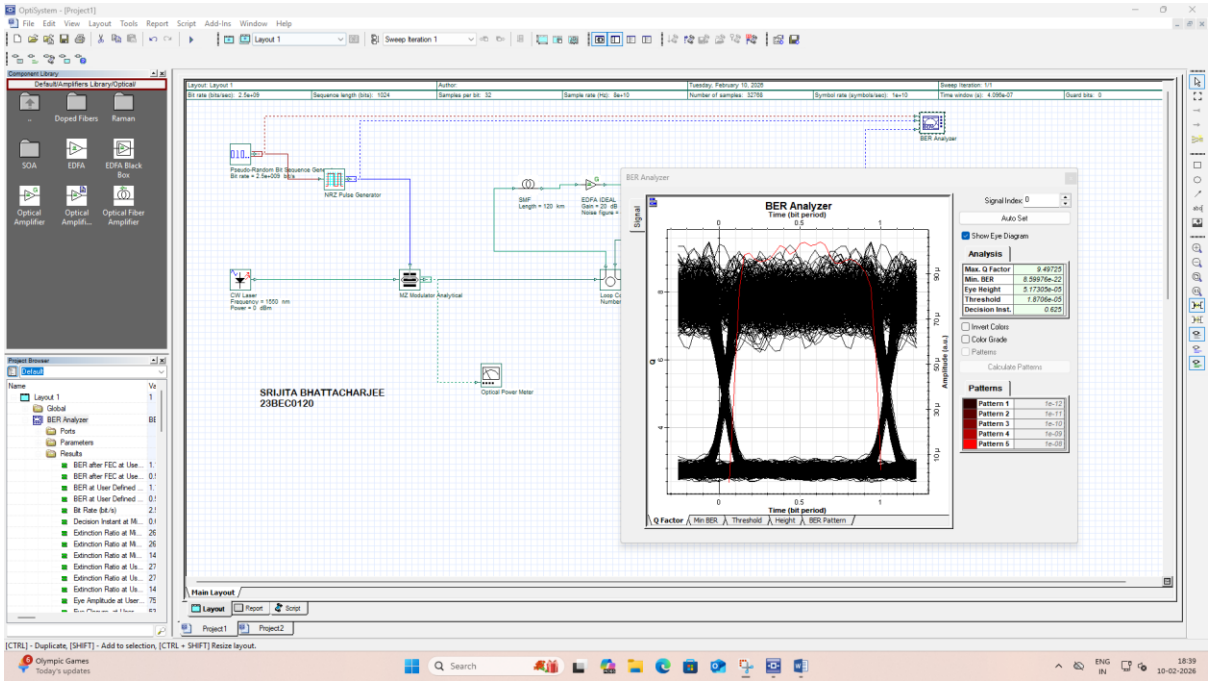
Result and Inference:

2.5 GBPS

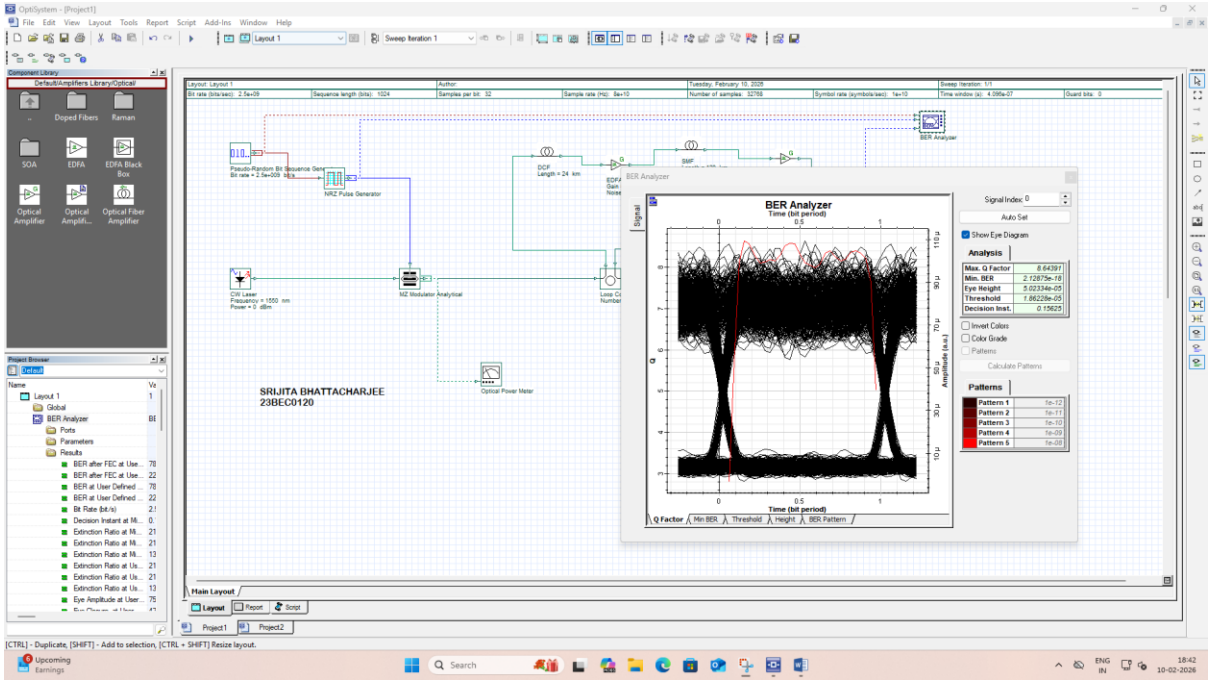
BEFORE COMPENSATION



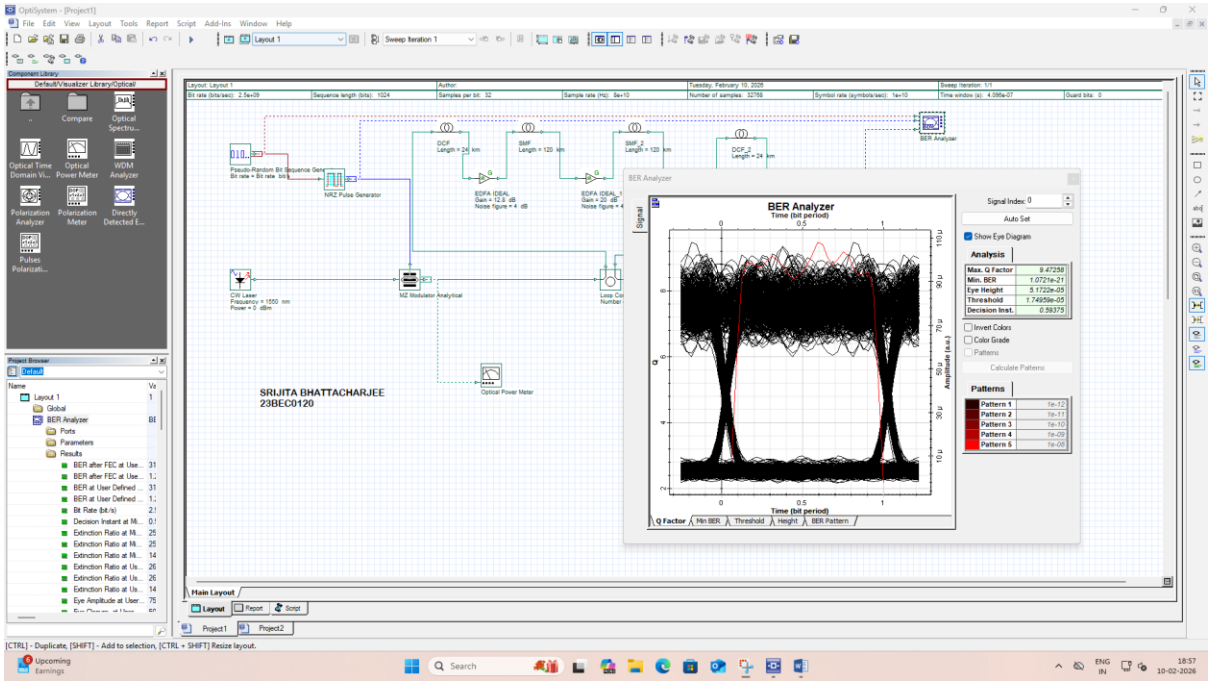
POST COMPENSATION



PRE COMPENSATION

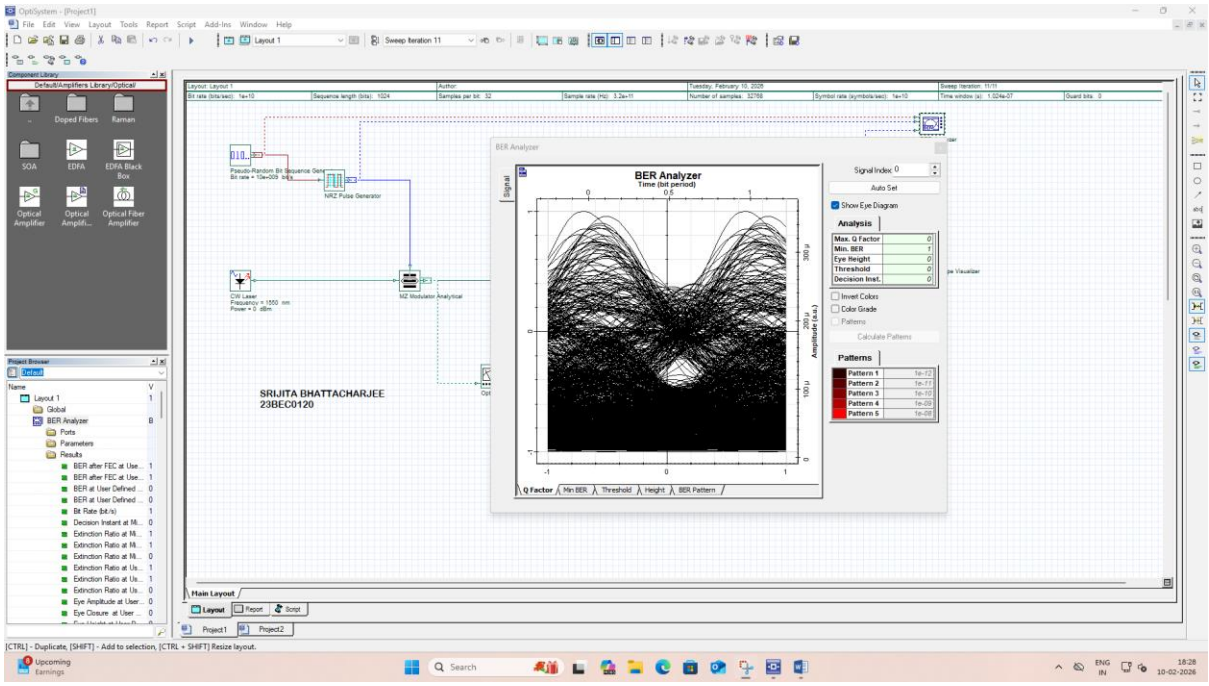


HYBRID COMPENSATION

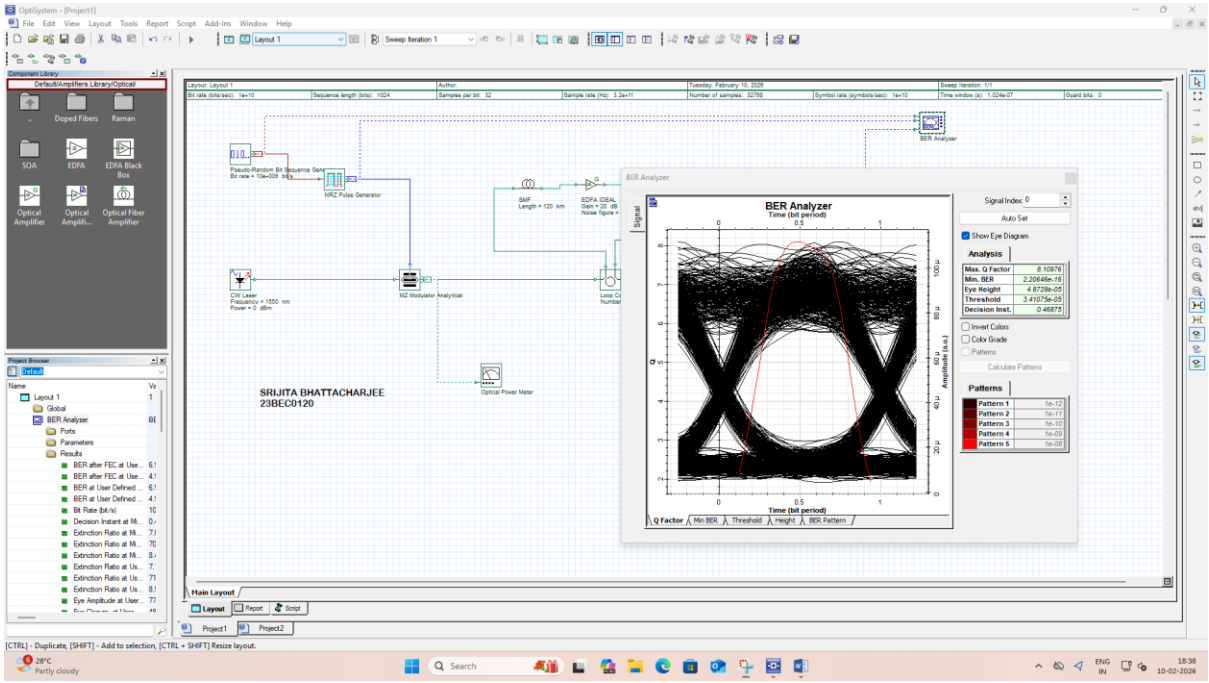


<u>10 GBPS</u>

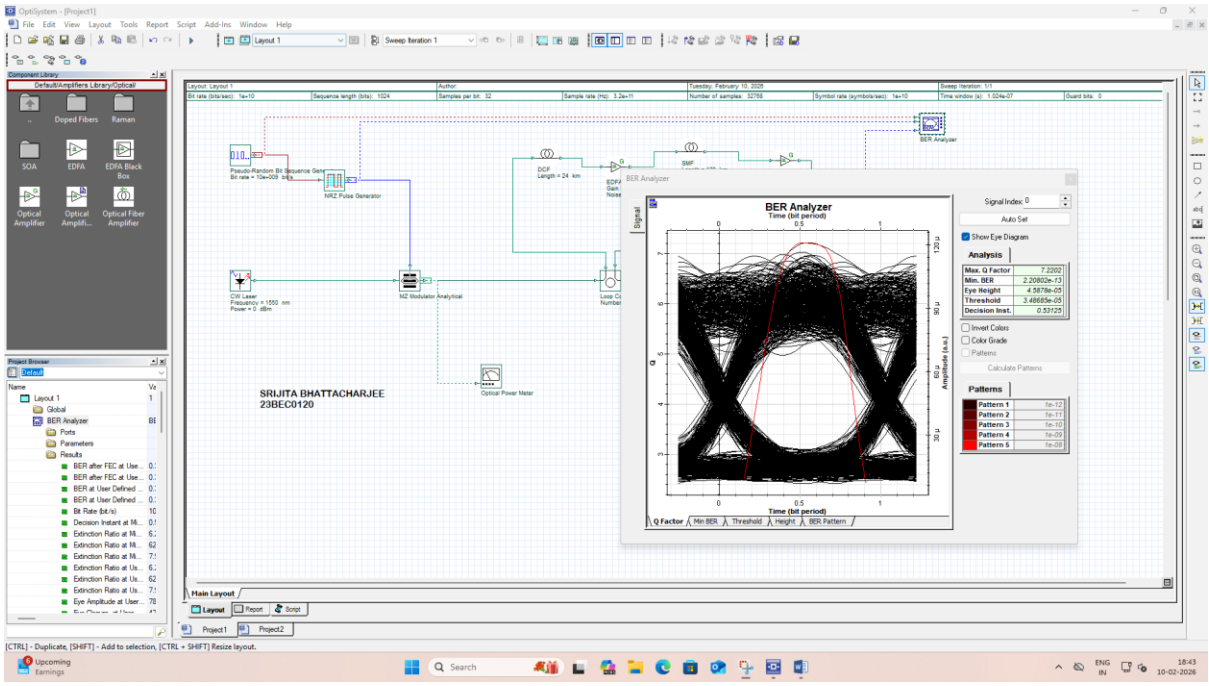
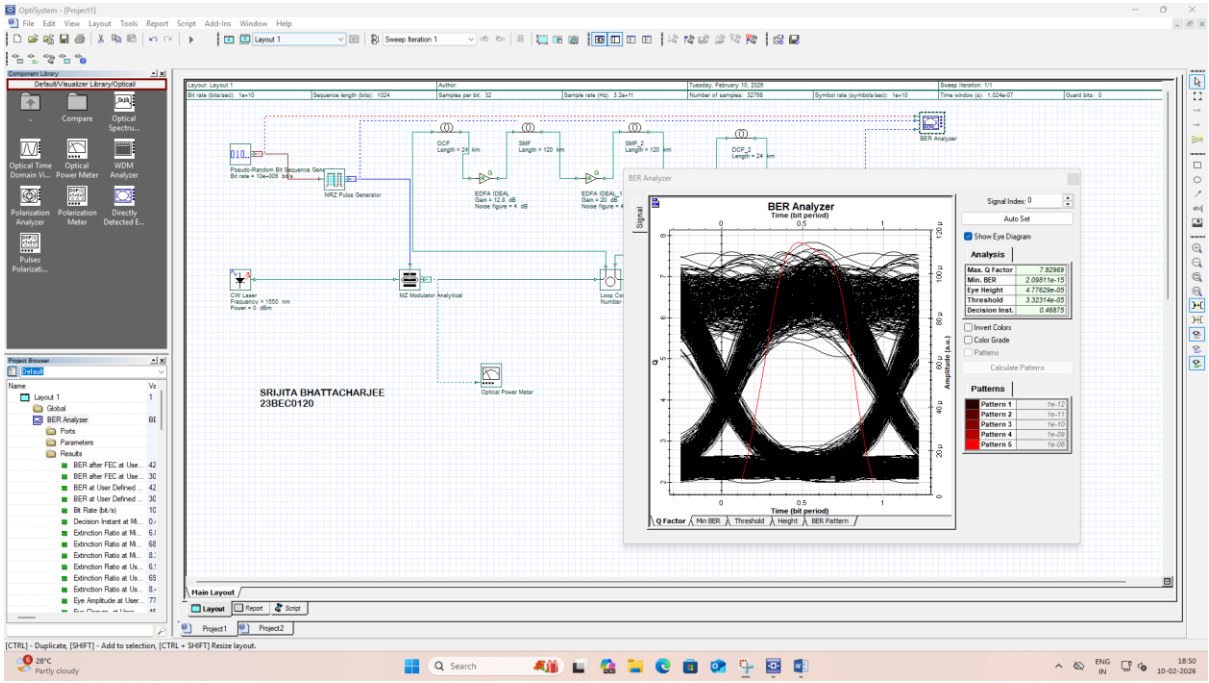
BEFORE COMPENSATION



POST COMPENSATION



PRE COMPENSATION



HYBRID COMPENSATION